

THEORETICAL EXISTENCE IN BANACH SPACES AND NUMERICAL SIMULATION OF A FIVE-COMPARTMENT FRACTIONAL-ORDER SMOKING MODEL

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Abstract

Smoking remains a significant public health burden whose population-level dynamics are inadequately captured by classical integer-order compartmental models, which lack the capacity to encode historical dependence in behavioral transitions. Using the Caputo–Fabrizio (CF) derivative, we propose and analyze a five-compartment fractional-order smoking model, which includes potential (P), occasional (I), heavy (H), quitting (Q), and recovered (R) individuals. Using the Banach contraction principle and Picard iteration, the existence and uniqueness of solutions in a Banach function space are established, and Ulam–Hyers stability is verified. To simulate fractional orders numerically, we develop a two-step Adams–Bashforth scheme applied to the CF integral. Finally, to recover biologically consistent parameter values, a Physics-Informed Neural

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Network (PINN) framework is constructed that approximates the five model states and estimates all ten biological parameters by minimising a composite loss enforcing the CF fractional equations, initial conditions, and data fidelity against CDC National Health Interview Survey cigarette smoking prevalence records for 2000–2023. Based on the estimated fractional order $\alpha = 0.95$, memory effects are measurable in smoking dynamics and in the recovered parameter values. The findings show that smoking initiation and preventive interventions dominate initial dynamics, whereas cessation and recovery parameters determine long-term asymptotic behavior. In the results, decreasing α leads to a decline of the potential smoker class, reduces the infection peak, and increases the recovery rate.

Keywords: Caputo–Fabrizio Fractional Derivative; Existence and Uniqueness; Smoking Dynamics; Physics-Informed Neural Networks; Parameter Estimation; Adams–Bashforth–Moulton Scheme; Ulam–Hyers Stability.

1. INTRODUCTION

Smoking tobacco is among the gravest and most enduring threats to global health. Approximately seven million people die from direct tobacco use each year, with the remainder dying from second-hand smoke, according to the World Health Organization. Worldwide, tobacco consumption results in an estimated one trillion dollars in direct health-care costs and productivity losses each year.^[12] In the United States, smoking is still the leading cause of preventable morbidity and mortality, resulting in over 480,000 deaths each year, 300 billion in medical costs, and lost productivity, according to the Centers for Disease Control and Prevention.^[3] The most recent CDC National Health Interview Survey data indicated that, despite decades of concerted public health efforts, such as restrictive advertising restrictions, progressive taxation, smoke-free workplace laws, and nationwide cessation campaigns, approximately 11.6% of US adults will continue to smoke cigarettes by the year 2022, highlighting the persistent challenge of achieving sustained population-wide smoking reductions.^[34] Physiologically addictive nicotine, coupled with deep-rooted social, cultural, and psychological determinants of smoking behavior, leads to low voluntary quitting rates among former smokers, and persistently high relapse rates, complicating intervention design and evaluation.^[5]

For understanding the dynamics of tobacco consumption at the population level, modeling has emerged as a powerful tool. A compartmental smoking model is based on classical epidemic theory, which views smoking as an “infectious” habit transmitted through social contact, with individuals progressing through a succession of well-defined stages

that include initiation, habituation, and cessation of smoking.^[67] Smoking is an important risk factor for respiratory infections, particularly pneumococcal pneumonia.^[8] Several refinements have been made since then, distinguishing occasional and heavy smokers, quitters, and relapsed former smokers, capturing the nonlinear interaction between transmission, progression, and recovery relevant to real smoking outcomes.^[9–11] For evaluating awareness campaigns, pharmacotherapy, and combined cessation strategies, optimal control approaches have been incorporated into these frameworks,^[12–13] while stochastic extensions have been employed to account for demographic variability in small populations. For quantifying smoking reproduction numbers, identifying the threshold conditions that separate smoking-free and smoking-endemic equilibrium, and determining long-term prevalence sensitivity to transmission and cessation parameters, smoking epidemic-style models have proven particularly useful.^[79–11] However, their reliance on integer-order ordinary differential equations imposes structural limitations that restrict their descriptive ability.

It is implicitly assumed that the instantaneous value of a system state determines the rate of state change in classical integer-order differential equation models. Smoking behavior is characterized by accumulation effects and nicotine-driven neuroadaptation, as well as a strong dependence on addiction history that lasts for months or years.^[17–19] A former smoker’s probability of relapse, for instance, is substantially conditioned on the duration and intensity of prior smoking episodes, a hereditary property that integer-order formulations cannot encode.^[18] Similarly, the speed at which an occasional smoker escalates to habitual

or heavy smoking is influenced by a cumulative social exposure history that the standard derivative fails to represent.^{[14][15]} A growing body of research applies fractional-order differential equations to epidemiological and biological systems in which memory effects and nonlocal temporal interactions are intrinsic characteristics of the underlying processes in part because Markovian models are inadequate in capturing such path-dependent dynamics.^{[16][18][19]}

For modeling systems with memory and hereditary properties, fractional calculus provides a natural and mathematically rigorous approach.^{[17][19]} The Caputo derivative^[14] is among the most widely used fractional operators because it permits conventional initial conditions and reduces to the standard derivative. Power-law kernels exhibit a non-integrable singularity at the origin, which can introduce numerical stiffness and restrict their applicability to biological systems.^[15] Caputo and Fabrizio^[15] developed a new fractional operator whose kernel is a smooth, non-singular exponential function that eliminates the singularity while preserving the essential nonlocality. Additionally, Atangana and Baleanu^[16] provide a generalization employing a Mittag-Leffler kernel to capture both exponential decay and algebraic tail behavior. Although each operator has distinct mathematical characteristics and ranges of applicability, fractional models consistently outperform their integer-order counterparts in reproducing the nonlocal, long-memory dynamics observed in ecological, epidemiological, and pharmacokinetic systems.^{[16][18][19]} The key advantage is the capacity of the fractional order α to serve as a continuous interpolation parameter that tunes the degree of memory in the system, enabling a closer calibration to empirical data than the binary choice between a first-order and a second-order integer derivative would allow.

The Caputo–Fabrizio (CF) derivative stands out among available fractional operators for several reasons that make it particularly well-suited to modeling smoking dynamics. First, its exponential kernel is non-singular and bounded on any finite interval, which simplifies both the theoretical analysis and the numerical implementation of the associated integral equations.^{[15][20]} Second, the corresponding integral operator takes a compact, closed form that combines a local instantaneous term and a weighted running average of the past trajectory, providing an interpretable decomposition of memory effects.^{[20][21]} Third, the CF operator maintains the semigroup property in an approximate sense

and possesses an associated Laplace transform of standard form, facilitating frequency-domain analysis and stability investigations.^{[21][22]} Fourth, numerical schemes constructed from the CF integral exhibit improved stability compared with those based on power-law kernels, which is particularly advantageous when solving coupled nonlinear systems over long time horizons, as arise naturally in multi-compartment smoking models.^{[23][24]} For these reasons, the CF derivative has attracted rapidly growing interest in applied mathematical modeling, and its deployment in smoking dynamics constitutes a theoretically motivated and computationally practical choice.

Recent literature has examined fractional-order extensions to smoking and epidemiological models. Rahman *et al.*^[25] analysed a fractional smoking model under the Caputo operator and established stability conditions for the smoking-free equilibrium, while Almutairi *et al.*^[26] compared integer and fractional formulations for a smoking cessation model and demonstrated the superior data-fitting capability of the fractional variant. Cattani *et al.*^[27] studied an Atangana–Baleanu fractional smoking model and derived existence and uniqueness conditions via a fixed-point argument. Fractional epidemic models have similarly proliferated across a wide range of infectious disease contexts.^{[28][31]} applied the CF derivative to compartmental model and obtained numerical solutions via the Adams–Bashforth scheme, while Khan *et al.*^[32] employed the CF operator in a hepatitis B transmission model and confirmed improved agreement with clinical prevalence records. The existence and stability of fractional differential systems in Banach spaces involving the Atangana–Baleanu and Caputo–Fabrizio operators have been extensively investigated (see Refs. [33] and [34]). Almeida *et al.*^[35] investigated the influence of the fractional order on the dynamics of an SIR epidemic model and found that varying α systematically shifts the timing and magnitude of the epidemic peak, a finding with direct relevance to intervention scheduling. Sensitivity analyses within fractional frameworks have been conducted by Huo *et al.*^[36] for a fractional model and by Zakirullah *et al.*^[37] for a generalized CF epidemic system, revealing that transmission and cessation parameters exert the dominant influence on long-term smoking and disease prevalence, respectively. However, a research gap remains: no existing study has combined, within a single coherent framework, the CF fractional formulation of

a multi-compartment smoking model, a complete suite of mathematical well-posedness results positivity, boundedness, existence, uniqueness, and Ulam–Hyers stability, an Adams–Bashforth numerical scheme adapted to the CF integral, comprehensive three-dimensional parameter sensitivity analyses, and machine-learning-based parameter estimation validated against real surveillance data. The present work addresses this gap in full.

Raissi *et al.*,^[38] in their modern form, introduced physics-informed neural networks to the numerical solution of differential equations. With PINNs, the governing equations are encoded in the loss function of a deep neural network as soft constraints rather than discretizing them on a computational mesh. A composite objective that balances data fidelity, equation residuals, and initial or boundary conditions can be minimized simultaneously to optimize the network weights as well as any unknown physical parameters.^{[38][39]} Physic residuals act as a principled regularizer that prevents overfitting to sparse data, making this architecture especially useful for inverse problems involving unknown parameter recovery from partial or noisy observations.^[40] PINNs have been applied to parameter identification in compartmental epidemic models by Cai *et al.*,^[41] who demonstrated accurate recovery of transmission and recovery rates for a COVID-19 SEIR system from limited case count data, and by Shaier *et al.*,^[42] who extended the approach to age-structured population dynamics. The application of PINNs to fractional differential equations has been demonstrated by Pang *et al.*,^[43] who showed that the CF fractional derivative can be incorporated into the PINN residual through a discrete convolution approximation derived from automatic differentiation. Given that real smoking prevalence data are available only as coarse annual aggregates and that the individual compartments of a five-class smoking model cannot be independently observed from standard surveillance instruments, the PINN framework offers an ideally suited methodology for joint state reconstruction and parameter estimation in the present context.

The present study makes the following original contributions to the mathematical modeling of smoking dynamics. Five-compartment CF fractional-order smoking models are proposed with an awareness parameter, in which the non-singular exponential kernel of the CF operator captures the memory-dependent nature of smoking initiation, relapse, and recovery that integer-order

formulations cannot. The positivity and boundedness of all solutions are proved within a forward-invariant feasible region, and existence and uniqueness are established via the Banach contraction principle and Picard iteration applied to the equivalent CF Volterra integral system, with an explicit contractivity condition on the kernel Lipschitz constants. Ulam–Hyers stability is verified under the same condition, confirming that small perturbations to the model produce proportionally bounded deviations in the solution trajectories. A two-step Adams–Bashforth scheme is derived from the CF integral via piecewise linear Lagrange interpolation, with rigorous local truncation error bounds, and used to produce time-domain simulations and phase portraits across multiple fractional orders. Finally, a PINN framework that jointly reconstructs all five compartment states and estimates all ten biological parameters is calibrated against CDC NHIS^[3] prevalence data and the CF fractional smoking model validated against national surveillance data via PINN-based inverse estimation.

This paper is organized as follows. Section 2 presents the preliminary definitions. Section 3 formulates the model and establishes positivity and boundedness of solutions. Section 4 develops the existence and uniqueness results via the Banach fixed-point theorem together with the Ulam–Hyers stability analysis. Section 6 develops the PINN framework for parameter estimation against CDC NHIS data, and Sec. 7 reports the numerical simulations. Finally, Sec. 8 concludes the paper.

2. DEFINITIONS AND PRELIMINARY CONCEPTS

In this section, we briefly recall the basic definition of the Caputo–Fabrizio (CF) fractional derivative. Further details can be found in Refs. [15] and [20].

Definition 2.1. Let $f \in H^1(a, b)$, where $b > a$, and let $\alpha \in (0, 1)$. The CF fractional derivative of order α is defined by

$$\begin{aligned} {}^{\text{CF}}D_t^\alpha f(t) &= \frac{M(\alpha)}{1-\alpha} \int_a^t f'(\tau) \\ &\quad \times \exp\left(-\frac{\alpha}{1-\alpha}(t-\tau)\right) d\tau, \quad t > a, \end{aligned} \quad (1)$$

where $M(\alpha)$ is a normalization function satisfying

$$M(0) = M(1) = 1.$$

Table 1 Nomenclature summary.

State Variables			
P	Potential smokers	Q	Quitters
I	Occasional smokers	R	Recovered
H	Heavy smokers	N	Total population
Biological Parameters			
Λ	Recruitment rate	ρ	Awareness/prevention rate
μ	Natural mortality	ρ_1, ρ_2	Quitting rates (I, H)
β, η	Initiation rate, heavy-smoker modifier	γ	Relapse rate
σ	Occasional→heavy rate	ω	Recovery rate
δ	Smoking-attributable mortality		
Fractional-operator & analysis Symbols			
α	CF fractional order	μ_i, μ^*	Kernel Lipschitz constants
${}^{\text{CF}}D^\alpha$	CF derivative operator	η	Ulam–Hyers stability constant
$\Upsilon_i(t, \mathbf{u})$	Compartment RHS kernel	L, L_i	Joint/component Lipschitz constants
Θ, θ	PINN parameters, weights		

The associated fractional integral is defined by

$${}^{\text{CF}}I_t^\alpha f(t) = \frac{2(1-\alpha)}{(2-\alpha)M(\alpha)} f(t) + \frac{2\alpha}{(2-\alpha)M(\alpha)} \int_a^t f(\tau) d\tau. \quad (2)$$

Since the fractional integral is required to be an average of $f(t)$ and its first-order integral, its coefficients satisfy

$$\frac{2(1-\alpha)}{(2-\alpha)M(\alpha)} + \frac{2\alpha}{(2-\alpha)M(\alpha)} = 1.$$

Hence, the normalization function is taken as

$$M(\alpha) = \frac{2}{2-\alpha}, \quad 0 \leq \alpha \leq 1. \quad (3)$$

Substituting (3) into the general integral form (2) gives the following closed-form expression for the associated CF fractional integral.

$${}^{\text{CF}}I_t^\alpha f(t) = (1-\alpha)f(t) + \alpha \int_a^t f(\tau) d\tau. \quad (4)$$

Remark 2.2. The kernel

$$\exp\left(-\frac{\alpha}{1-\alpha}(t-\tau)\right)$$

is non-singular at $\tau = t$ and accounts for the memory effect in the CF derivative.

Remark 2.3. If $f \notin H^1(a, b)$, the CF derivative can alternatively be represented in terms of the function

values as

$${}^{\text{CF}}D_t^\alpha f(t) = \frac{\alpha M(\alpha)}{1-\alpha} \int_a^t [f(t) - f(\tau)] \times \exp\left(-\frac{\alpha}{1-\alpha}(t-\tau)\right) d\tau, \quad (5)$$

whenever the corresponding integral is well defined.

Lemma 2.4. Let $x : (0, T) \rightarrow \mathbb{R}$ satisfy

$${}^{\text{CF}}D_t^\alpha x(t) \leq -\mu x(t) + h(t)$$

for some $\mu > 0$, where h is bounded and non-negative. Then $x(t)$ is bounded for all $t > 0$.

Nomenclature

All symbols throughout this work are defined in Table 1.

3. MODEL FORMULATION

In time t , the total population is divided into five compartments based on smoking status, namely potential smokers $P(t)$, occasional smokers $I(t)$, heavy smokers $H(t)$, quitters $Q(t)$, and permanently recovered individuals $R(t)$. Accordingly, the total population satisfies

$$N(t) = P(t) + I(t) + H(t) + Q(t) + R(t).$$

In a population, new individuals are recruited at a constant rate of Λ , and all individuals are subject to natural mortality at μ . Potential smokers become

occasional smokers through contact with existing smokers at transmission rate β , where heavy smokers contribute proportionally through the modification parameter η . Occasional smokers progress to heavy smoking at rate σ , whereas smoking-related mortality occurs among heavy smokers at rate δ . Individuals may quit smoking from the occasional and heavy smoker classes at rates ρ_1 and ρ_2 , respectively, entering the quitter compartment. Quitters may relapse to the potential smoker class at rate γ or achieve permanent recovery at rate ω . In addition, awareness and preventive intervention programs reduce the number of potential smokers through the control parameter ρ .

Classical integer-order smoking models are governed by the following ordinary differential equations in accordance with these biological assumptions:

$$\begin{aligned}\frac{dP}{dt} &= \Lambda - \beta P(I + \eta H) - (\mu + \rho)P + \gamma Q, \\ \frac{dI}{dt} &= \beta P(I + \eta H) - (\mu + \sigma + \rho_1)I, \\ \frac{dH}{dt} &= \sigma I - (\mu + \delta + \rho_2)H, \\ \frac{dQ}{dt} &= \rho_1 I + \rho_2 H - (\mu + \gamma + \omega)Q, \\ \frac{dR}{dt} &= \omega Q - \mu R.\end{aligned}\tag{6}$$

Although the classical model describes the main smoking transitions, smoking initiation, dependence, relapse, and recovery are influenced by previous behavioral history and accumulated exposure. To account for these memory-dependent effects, we replace the integer-order derivative with the CF derivative of order $0 < \alpha \leq 1$, which incorporates memory through its non-singular kernel.

Consequently, the proposed fractional-order smoking model is formulated as

$$\begin{aligned}{}^{\text{CF}}D_t^\alpha P &= \Lambda - \beta P(I + \eta H) - (\mu + \rho)P + \gamma Q, \\ {}^{\text{CF}}D_t^\alpha I &= \beta P(I + \eta H) - (\mu + \sigma + \rho_1)I, \\ {}^{\text{CF}}D_t^\alpha H &= \sigma I - (\mu + \delta + \rho_2)H, \\ {}^{\text{CF}}D_t^\alpha Q &= \rho_1 I + \rho_2 H - (\mu + \gamma + \omega)Q, \\ {}^{\text{CF}}D_t^\alpha R &= \omega Q - \mu R.\end{aligned}\tag{7}$$

3.1. Positivity and Boundedness of Solutions

For the fractional-order smoking model to be epidemiologically meaningful, all compartment variables must remain non-negative and bounded for all $t \geq 0$. The following theorem establishes these properties simultaneously.

Theorem 3.1. *Suppose that the initial conditions of system (7) satisfy*

$$\begin{aligned}P(0) &\geq 0, \quad I(0) \geq 0, \quad H(0) \geq 0, \\ Q(0) &\geq 0, \quad R(0) \geq 0.\end{aligned}$$

Then the solution of system (7) remains non-negative and bounded for all $t \geq 0$. In particular,

$$(P(t), I(t), H(t), Q(t), R(t)) \in \mathbb{R}_+^5, \quad t \geq 0$$

and the total population

$$N(t) = P(t) + I(t) + H(t) + Q(t) + R(t)$$

satisfies

$$N(t) \leq \frac{\Lambda}{\mu}$$

asymptotically. Consequently, the positively invariant feasible region may be taken as

$$\begin{aligned}\Omega = \left\{ (P, I, H, Q, R) \in \mathbb{R}_+^5 : \right. \\ \left. P + I + H + Q + R \leq \frac{\Lambda}{\mu} \right\}.\end{aligned}\tag{8}$$

Proof. We first establish positivity. Consider the boundary of the positive orthant $\mathbb{R}_+^5$. From system (7), we have

$$\begin{aligned}{}^{\text{CF}}D_t^\alpha P|_{P=0} &= \Lambda + \gamma Q \geq 0, \\ {}^{\text{CF}}D_t^\alpha I|_{I=0} &= \beta P \eta H \geq 0, \\ {}^{\text{CF}}D_t^\alpha H|_{H=0} &= \sigma I \geq 0, \\ {}^{\text{CF}}D_t^\alpha Q|_{Q=0} &= \rho_1 I + \rho_2 H \geq 0, \\ {}^{\text{CF}}D_t^\alpha R|_{R=0} &= \omega Q \geq 0.\end{aligned}$$

Thus, the vector field is directed into the positive orthant on every boundary face. Hence, solutions that start in $\mathbb{R}_+^5$ cannot leave this region, and therefore

$$P(t), I(t), H(t), Q(t), R(t) \geq 0, \quad t \geq 0.$$

Next, define the total population by

$$N(t) = P(t) + I(t) + H(t) + Q(t) + R(t).$$

Using the linearity of the CF derivative and summing the five equations in (7), we obtain

$${}^{\text{CF}}D_t^\alpha N(t) = \Lambda - \mu N(t) - \rho P(t) - \delta H(t). \quad (9)$$

Since $P(t), H(t) \geq 0$ and $\rho, \delta \geq 0$, it follows that

$${}^{\text{CF}}D_t^\alpha N(t) \leq \Lambda - \mu N(t). \quad (10)$$

Therefore, the total population is ultimately bounded, with

$$\limsup_{t \rightarrow \infty} N(t) \leq \frac{\Lambda}{\mu}.$$

Hence, all five non-negative compartment variables are ultimately bounded. Consequently, the feasible region (8) provides a biologically meaningful bounded region for system (7). $\square$

4. EXISTENCE AND UNIQUENESS OF SOLUTIONS

We establish the existence and uniqueness of solutions of the fractional smoking model by applying the Banach fixed-point theorem. Let $t_0 > 0$ and consider the Banach space

$$\mathcal{X} = C([0, t_0], \mathbb{R}^5), \quad (11)$$

equipped with the supremum norm

$$\begin{aligned} \|\mathbf{u}\|_{\mathcal{X}} &= \sup_{t \in [0, t_0]} \|\mathbf{u}(t)\|_{\infty}, \\ \mathbf{u}(t) &= (P(t), I(t), H(t), Q(t), R(t))^{\top}. \end{aligned} \quad (12)$$

For a fixed initial state

$$\mathbf{u}_0 = (P(0), I(0), H(0), Q(0), R(0))^{\top},$$

we define the closed ball

$$\mathcal{B}_r = \{\mathbf{u} \in \mathcal{X} : \|\mathbf{u} - \mathbf{u}_0\|_{\mathcal{X}} \leq r\}, \quad (13)$$

for some $r > 0$.

Using the CF fractional integral specified in Sec. 2, system (7) can be written as the equivalent Volterra integral system

$$\begin{aligned} P(t) &= P(0) + (1 - \alpha)\Upsilon_1(t, \mathbf{u}) \\ &\quad + \alpha \int_0^t \Upsilon_1(s, \mathbf{u}(s)) \, ds, \\ I(t) &= I(0) + (1 - \alpha)\Upsilon_2(t, \mathbf{u}) \\ &\quad + \alpha \int_0^t \Upsilon_2(s, \mathbf{u}(s)) \, ds, \end{aligned}$$

$$\begin{aligned} H(t) &= H(0) + (1 - \alpha)\Upsilon_3(t, \mathbf{u}) \\ &\quad + \alpha \int_0^t \Upsilon_3(s, \mathbf{u}(s)) \, ds, \\ Q(t) &= Q(0) + (1 - \alpha)\Upsilon_4(t, \mathbf{u}) \\ &\quad + \alpha \int_0^t \Upsilon_4(s, \mathbf{u}(s)) \, ds, \\ R(t) &= R(0) + (1 - \alpha)\Upsilon_5(t, \mathbf{u}) \\ &\quad + \alpha \int_0^t \Upsilon_5(s, \mathbf{u}(s)) \, ds, \end{aligned} \quad (14)$$

where

$$\begin{aligned} \Upsilon_1(t, \mathbf{u}) &= \Lambda - \beta P(I + \eta H) \\ &\quad - (\mu + \rho)P + \gamma Q, \\ \Upsilon_2(t, \mathbf{u}) &= \beta P(I + \eta H) - (\mu + \sigma + \rho_1)I, \\ \Upsilon_3(t, \mathbf{u}) &= \sigma I - (\mu + \delta + \rho_2)H, \\ \Upsilon_4(t, \mathbf{u}) &= \rho_1 I + \rho_2 H - (\mu + \gamma + \omega)Q, \\ \Upsilon_5(t, \mathbf{u}) &= \omega Q - \mu R. \end{aligned} \quad (15)$$

Theorem 4.1. Suppose that the components of $\mathbf{u}, \mathbf{v} \in \mathcal{B}_r$ satisfy

$$\begin{aligned} |P(t)|, |\tilde{P}(t)| &\leq M_P, \quad |I(t)|, |\tilde{I}(t)| \leq M_I, \\ |H(t)|, |\tilde{H}(t)| &\leq M_H, \quad |Q(t)|, |\tilde{Q}(t)| \leq M_Q, \\ |R(t)|, |\tilde{R}(t)| &\leq M_R, \end{aligned}$$

for all $t \in [0, t_0]$, where $\mathbf{u} = (P, I, H, Q, R)^{\top}$ and $\mathbf{v} = (\tilde{P}, \tilde{I}, \tilde{H}, \tilde{Q}, \tilde{R})^{\top}$. Then the vector-valued mapping

$$\Upsilon = (\Upsilon_1, \Upsilon_2, \Upsilon_3, \Upsilon_4, \Upsilon_5)^{\top}$$

is Lipschitz continuous on $\mathcal{B}_r$ with respect to the ℓ_{∞} norm. In particular,

$$\|\Upsilon(t, \mathbf{u}) - \Upsilon(t, \mathbf{v})\|_{\infty} \leq L \|\mathbf{u} - \mathbf{v}\|_{\infty}, \quad (16)$$

where

$$L = \max\{L_1, L_2, L_3, L_4, L_5\} \quad (17)$$

and

$$\begin{aligned} L_1 &= \beta(M_I + \eta M_H) + (\mu + \rho) + \beta M_P \\ &\quad + \beta \eta M_P + \gamma, \\ L_2 &= \beta(M_I + \eta M_H) + \beta M_P + \beta \eta M_P \\ &\quad + \mu + \sigma + \rho_1, \end{aligned}$$

$$\begin{aligned} L_3 &= \sigma + \mu + \delta + \rho_2, \\ L_4 &= \rho_1 + \rho_2 + \mu + \gamma + \omega, \\ L_5 &= \omega + \mu. \end{aligned} \quad (18)$$

Proof. Let

$$\begin{aligned} \mathbf{u} &= (P, I, H, Q, R)^\top, \\ \mathbf{v} &= (\tilde{P}, \tilde{I}, \tilde{H}, \tilde{Q}, \tilde{R})^\top \end{aligned}$$

belong to $\mathcal{B}_r$. For the first component, we have

$$\begin{aligned} &|\Upsilon_1(t, \mathbf{u}) - \Upsilon_1(t, \mathbf{v})| \\ &= |-\beta[P(I + \eta H) - \tilde{P}(\tilde{I} + \eta \tilde{H})] \\ &\quad - (\mu + \rho)(P - \tilde{P}) + \gamma(Q - \tilde{Q})|. \end{aligned} \quad (19)$$

Using

$$\begin{aligned} &P(I + \eta H) - \tilde{P}(\tilde{I} + \eta \tilde{H}) \\ &= (I + \eta H)(P - \tilde{P}) + \tilde{P}(I - \tilde{I}) \\ &\quad + \eta \tilde{P}(H - \tilde{H}), \end{aligned}$$

together with the bounds on $\mathcal{B}_r$, we obtain

$$\begin{aligned} &|\Upsilon_1(t, \mathbf{u}) - \Upsilon_1(t, \mathbf{v})| \\ &\leq [\beta(M_I + \eta M_H) + (\mu + \rho)]|P - \tilde{P}| \\ &\quad + \beta M_P|I - \tilde{I}| + \beta \eta M_P|H - \tilde{H}| + \gamma|Q - \tilde{Q}| \\ &\leq L_1 \|\mathbf{u} - \mathbf{v}\|_\infty. \end{aligned} \quad (20)$$

Similarly,

$$\begin{aligned} &|\Upsilon_2(t, \mathbf{u}) - \Upsilon_2(t, \mathbf{v})| \leq L_2 \|\mathbf{u} - \mathbf{v}\|_\infty, \\ &|\Upsilon_3(t, \mathbf{u}) - \Upsilon_3(t, \mathbf{v})| \leq L_3 \|\mathbf{u} - \mathbf{v}\|_\infty, \\ &|\Upsilon_4(t, \mathbf{u}) - \Upsilon_4(t, \mathbf{v})| \leq L_4 \|\mathbf{u} - \mathbf{v}\|_\infty, \\ &|\Upsilon_5(t, \mathbf{u}) - \Upsilon_5(t, \mathbf{v})| \leq L_5 \|\mathbf{u} - \mathbf{v}\|_\infty. \end{aligned} \quad (21)$$

Taking the ℓ_∞ maximum over the five component bounds (20)–(21) gives

$$\|\Upsilon(t, \mathbf{u}) - \Upsilon(t, \mathbf{v})\|_\infty \leq L \|\mathbf{u} - \mathbf{v}\|_\infty,$$

with L defined by (17). Hence, Υ is Lipschitz continuous on $\mathcal{B}_r$. $\square$

To construct the solution, define the Picard sequence

$$\mathbf{u}^{(0)}(t) = \mathbf{u}_0$$

and, for $n \geq 1$, let

$$\begin{aligned} \mathbf{u}^{(n)}(t) &= \mathbf{u}_0 + (1 - \alpha)\Upsilon(t, \mathbf{u}^{(n-1)}(t)) \\ &\quad + \alpha \int_0^t \Upsilon(s, \mathbf{u}^{(n-1)}(s)) ds. \end{aligned} \quad (22)$$

Define the operator $\mathcal{T} : \mathcal{B}_r \rightarrow \mathcal{X}$ by

$$\begin{aligned} (\mathcal{T}\mathbf{u})(t) &= \mathbf{u}_0 + (1 - \alpha)\Upsilon(t, \mathbf{u}(t)) \\ &\quad + \alpha \int_0^t \Upsilon(s, \mathbf{u}(s)) ds. \end{aligned} \quad (23)$$

For any $\mathbf{u}, \mathbf{v} \in \mathcal{B}_r$, Theorem 4.1 gives

$$\begin{aligned} &\|(\mathcal{T}\mathbf{u}) - (\mathcal{T}\mathbf{v})\|_{\mathcal{X}} \\ &\leq (1 - \alpha)\|\Upsilon(\cdot, \mathbf{u}) - \Upsilon(\cdot, \mathbf{v})\|_{\mathcal{X}} \\ &\quad + \alpha \sup_{t \in [0, t_0]} \int_0^t \|\Upsilon(s, \mathbf{u}) - \Upsilon(s, \mathbf{v})\|_\infty ds \\ &\leq [(1 - \alpha) + \alpha t_0]L \|\mathbf{u} - \mathbf{v}\|_{\mathcal{X}}. \end{aligned} \quad (24)$$

Define

$$\kappa_\alpha(t_0) = (1 - \alpha) + \alpha t_0. \quad (25)$$

Therefore, if

$$\kappa_\alpha(t_0)L < 1, \quad (26)$$

then $\mathcal{T}$ is a contraction on $\mathcal{B}_r$.

Theorem 4.2. The fractional smoking model (7) possesses a unique solution in $\mathcal{X}$ on the interval $[0, t_0]$ provided that

$$[(1 - \alpha) + \alpha t_0]L < 1, \quad (27)$$

where L is the joint Lipschitz constant defined by (17).

Proof. Let $\mathbf{u}$ and $\mathbf{v}$ be two solutions of (7) with the same initial condition. From (23),

$$\begin{aligned} \|\mathbf{u} - \mathbf{v}\|_{\mathcal{X}} &= \|\mathcal{T}\mathbf{u} - \mathcal{T}\mathbf{v}\|_{\mathcal{X}} \\ &\leq \kappa_\alpha(t_0)L \|\mathbf{u} - \mathbf{v}\|_{\mathcal{X}}. \end{aligned} \quad (28)$$

Consequently,

$$[1 - \kappa_\alpha(t_0)L] \|\mathbf{u} - \mathbf{v}\|_{\mathcal{X}} \leq 0. \quad (29)$$

Under condition (27), the factor $1 - \kappa_\alpha(t_0)L$ is strictly positive. Hence,

$$\|\mathbf{u} - \mathbf{v}\|_{\mathcal{X}} = 0,$$

which implies

$$\mathbf{u}(t) = \mathbf{v}(t), \quad t \in [0, t_0].$$

Thus, the solution is unique. Moreover, since $\mathcal{T}$ is a contraction on the complete space $\mathcal{B}_r$, the Banach fixed-point theorem guarantees the existence of a fixed point of $\mathcal{T}$, and therefore the existence of a solution. $\square$

4.1. Ulam–Hyers Stability

Ulam–Hyers (UH) stability describes the robustness of the smoking model (7) with respect to small perturbations in the governing equations. In particular, it guarantees that an approximate solution remains close to an exact solution when the perturbation is sufficiently small. Related existence and stability results for fractional systems in Banach spaces involving Atangana–Baleanu and Caputo–Fabrizio operators were established in Refs. [33] and [34]. Here, we extend this framework to the coupled five-compartment CF smoking model.

Let

$$\epsilon = (\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4, \epsilon_5)^\top \in \mathbb{R}_+^5.$$

The system (7) is said to be Ulam–Hyers stable on $[0, t_0]$ if, for every ϵ , each approximate trajectory $\mathbf{u}$ satisfying the perturbed system admits an exact solution $\mathbf{u}^*$ such that

$$\|\mathbf{u} - \mathbf{u}^*\|_{\mathcal{X}} \leq \eta \|\epsilon\|_{\infty}, \quad (30)$$

for some constant $\eta > 0$ independent of the particular approximate trajectory. Equivalently, the corresponding componentwise estimates are

$$\begin{aligned} \|P - P^*\|_{\infty} &\leq \eta \epsilon_1, & \|I - I^*\|_{\infty} &\leq \eta \epsilon_2, \\ \|H - H^*\|_{\infty} &\leq \eta \epsilon_3, & \|Q - Q^*\|_{\infty} &\leq \eta \epsilon_4, \\ \|R - R^*\|_{\infty} &\leq \eta \epsilon_5. \end{aligned} \quad (31)$$

Remark 4.1. Suppose that an approximate trajectory $\mathbf{u} = (P, I, H, Q, R)^\top$ satisfies

$$\|\text{CF}D_t^\alpha P(t) - \Upsilon_1(t, \mathbf{u})\| \leq \epsilon_1. \quad (32)$$

Then there exists a continuous perturbation function $h_1(t)$ such that

$$|h_1(t)| \leq \epsilon_1$$

and

$$\text{CF}D_t^\alpha P(t) = \Upsilon_1(t, \mathbf{u}) + h_1(t). \quad (33)$$

Analogously, there exist perturbation functions h_2, h_3, h_4, h_5 satisfying

$$|h_i(t)| \leq \epsilon_i, \quad i = 1, \dots, 5.$$

Thus, the perturbed vector system can be written as

$$\text{CF}D_t^\alpha \mathbf{u}(t) = \Upsilon(t, \mathbf{u}(t)) + \mathbf{h}(t), \quad (34)$$

where

$$\mathbf{h}(t) = (h_1(t), h_2(t), h_3(t), h_4(t), h_5(t))^\top$$

and

$$\|\mathbf{h}(t)\|_{\infty} \leq \|\epsilon\|_{\infty}.$$

Theorem 4.3. Assume that the joint Lipschitz condition (16) holds and that the contraction condition (27) is satisfied. Then the CF smoking system (7) is Ulam–Hyers stable on $[0, t_0]$. More precisely, the stability constant can be taken as

$$\eta = \frac{\kappa_\alpha(t_0)}{1 - \kappa_\alpha(t_0)L}, \quad \kappa_\alpha(t_0) = (1 - \alpha) + \alpha t_0, \quad (35)$$

where L is the joint Lipschitz constant defined in (17). Hence,

$$\|\mathbf{u} - \mathbf{u}^*\|_{\mathcal{X}} \leq \eta \|\epsilon\|_{\infty}. \quad (36)$$

Proof. Let $\mathbf{u}$ be an approximate solution satisfying

$$\text{CF}D_t^\alpha \mathbf{u}(t) = \Upsilon(t, \mathbf{u}(t)) + \mathbf{h}(t),$$

with

$$\|\mathbf{h}(t)\|_{\infty} \leq \|\epsilon\|_{\infty}.$$

Using the CF integral representation given in Sec. 2, the perturbed solution satisfies

$$\begin{aligned} \mathbf{u}(t) &= \mathbf{u}_0 + (1 - \alpha)\Upsilon(t, \mathbf{u}(t)) \\ &\quad + \alpha \int_0^t \Upsilon(s, \mathbf{u}(s)) ds \\ &\quad + (1 - \alpha)\mathbf{h}(t) + \alpha \int_0^t \mathbf{h}(s) ds. \end{aligned} \quad (37)$$

Let $\mathbf{u}^*$ denote the exact solution, which satisfies the same relation without the perturbation term. Subtracting the two relations and taking the ℓ_∞ norm gives

$$\begin{aligned} \|\mathbf{u} - \mathbf{u}^*\|_{\mathcal{X}} &\leq (1 - \alpha)\|\Upsilon(\cdot, \mathbf{u}) - \Upsilon(\cdot, \mathbf{u}^*)\|_{\mathcal{X}} \\ &\quad + \alpha \sup_{t \in [0, t_0]} \int_0^t \|\Upsilon(s, \mathbf{u}(s)) - \Upsilon(s, \mathbf{u}^*(s))\|_{\infty} ds \\ &\quad + (1 - \alpha)\|\epsilon\|_{\infty} + \alpha \sup_{t \in [0, t_0]} \int_0^t \|\epsilon\|_{\infty} ds. \end{aligned} \quad (38)$$

Using the joint Lipschitz condition (16), we obtain

$$\begin{aligned} \|\mathbf{u} - \mathbf{u}^*\|_{\mathcal{X}} &\leq [(1 - \alpha) + \alpha t_0]L \|\mathbf{u} - \mathbf{u}^*\|_{\mathcal{X}} \\ &\quad + [(1 - \alpha) + \alpha t_0]\|\epsilon\|_{\infty}. \end{aligned} \quad (39)$$

Introducing

$$\kappa_{\alpha}(t_0) = (1 - \alpha) + \alpha t_0,$$

we have

$$(1 - \kappa_{\alpha}(t_0)L)\|\mathbf{u} - \mathbf{u}^*\|_{\mathcal{X}} \leq \kappa_{\alpha}(t_0)\|\epsilon\|_{\infty}. \quad (40)$$

By the contraction condition (27),

$$1 - \kappa_{\alpha}(t_0)L > 0.$$

Therefore,

$$\|\mathbf{u} - \mathbf{u}^*\|_{\mathcal{X}} \leq \frac{\kappa_{\alpha}(t_0)}{1 - \kappa_{\alpha}(t_0)L}\|\epsilon\|_{\infty}, \quad (41)$$

which is precisely (36). Hence, the system is Ulam–Hyers stable on the finite interval $[0, t_0]$. $\square$

Remark 4.2. The UH stability constant is

$$\eta(\alpha, t_0) = \frac{(1 - \alpha) + \alpha t_0}{1 - [(1 - \alpha) + \alpha t_0]L},$$

with

$$\frac{\partial \kappa_{\alpha}(t_0)}{\partial \alpha} = t_0 - 1.$$

Hence, $\kappa_{\alpha}(t_0)$ increases with α for $t_0 > 1$ and decreases for $0 < t_0 < 1$. The condition $\kappa_{\alpha}(t_0)L < 1$ ensures that $\eta(\alpha, t_0)$ is finite and positive.

5. NUMERICAL SCHEME

Using the CF integral representation from Sec. 2, we construct a two-step Adams–Bashforth scheme for the numerical solution of the fractional smoking model. The formulation is based on the non-singular CF kernel and is written directly in terms of the model kernels Υ_i , $i = 1, \dots, 5$.

From the CF integral representation, system (7) can be written as

$$\begin{aligned} P(t) - P(0) &= (1 - \alpha)\Upsilon_1(t, \mathbf{u}(t)) \\ &\quad + \alpha \int_0^t \Upsilon_1(\tau, \mathbf{u}(\tau)) d\tau, \\ I(t) - I(0) &= (1 - \alpha)\Upsilon_2(t, \mathbf{u}(t)) \\ &\quad + \alpha \int_0^t \Upsilon_2(\tau, \mathbf{u}(\tau)) d\tau, \end{aligned}$$

$$\begin{aligned} H(t) - H(0) &= (1 - \alpha)\Upsilon_3(t, \mathbf{u}(t)) \\ &\quad + \alpha \int_0^t \Upsilon_3(\tau, \mathbf{u}(\tau)) d\tau, \\ Q(t) - Q(0) &= (1 - \alpha)\Upsilon_4(t, \mathbf{u}(t)) \\ &\quad + \alpha \int_0^t \Upsilon_4(\tau, \mathbf{u}(\tau)) d\tau, \\ R(t) - R(0) &= (1 - \alpha)\Upsilon_5(t, \mathbf{u}(t)) \\ &\quad + \alpha \int_0^t \Upsilon_5(\tau, \mathbf{u}(\tau)) d\tau. \end{aligned} \quad (42)$$

Let

$$t_n = nh, \quad n = 0, 1, \dots, N,$$

where $h > 0$ is the uniform step size. At $t = t_{n+1}$, we write

$$\begin{aligned} P_{n+1} - P_0 &= (1 - \alpha)\Upsilon_1(t_n, \mathbf{u}_n) \\ &\quad + \alpha \int_0^{t_{n+1}} \Upsilon_1(t, \mathbf{u}(t)) dt, \\ I_{n+1} - I_0 &= (1 - \alpha)\Upsilon_2(t_n, \mathbf{u}_n) \\ &\quad + \alpha \int_0^{t_{n+1}} \Upsilon_2(t, \mathbf{u}(t)) dt, \\ H_{n+1} - H_0 &= (1 - \alpha)\Upsilon_3(t_n, \mathbf{u}_n) \\ &\quad + \alpha \int_0^{t_{n+1}} \Upsilon_3(t, \mathbf{u}(t)) dt, \\ Q_{n+1} - Q_0 &= (1 - \alpha)\Upsilon_4(t_n, \mathbf{u}_n) \\ &\quad + \alpha \int_0^{t_{n+1}} \Upsilon_4(t, \mathbf{u}(t)) dt, \\ R_{n+1} - R_0 &= (1 - \alpha)\Upsilon_5(t_n, \mathbf{u}_n) \\ &\quad + \alpha \int_0^{t_{n+1}} \Upsilon_5(t, \mathbf{u}(t)) dt. \end{aligned} \quad (43)$$

Similarly, at $t = t_n$,

$$\begin{aligned} P_n - P_0 &= (1 - \alpha)\Upsilon_1(t_{n-1}, \mathbf{u}_{n-1}) \\ &\quad + \alpha \int_0^{t_n} \Upsilon_1(t, \mathbf{u}(t)) dt, \\ I_n - I_0 &= (1 - \alpha)\Upsilon_2(t_{n-1}, \mathbf{u}_{n-1}) \\ &\quad + \alpha \int_0^{t_n} \Upsilon_2(t, \mathbf{u}(t)) dt, \end{aligned}$$

$$\begin{aligned}
 H_n - H_0 &= (1 - \alpha) \Upsilon_3(t_{n-1}, \mathbf{u}_{n-1}) \\
 &\quad + \alpha \int_0^{t_n} \Upsilon_3(t, \mathbf{u}(t)) dt, \\
 Q_n - Q_0 &= (1 - \alpha) \Upsilon_4(t_{n-1}, \mathbf{u}_{n-1}) \\
 &\quad + \alpha \int_0^{t_n} \Upsilon_4(t, \mathbf{u}(t)) dt, \\
 R_n - R_0 &= (1 - \alpha) \Upsilon_5(t_{n-1}, \mathbf{u}_{n-1}) \\
 &\quad + \alpha \int_0^{t_n} \Upsilon_5(t, \mathbf{u}(t)) dt.
 \end{aligned} \tag{44}$$

Subtracting (44) from (43) gives

$$\begin{aligned}
 P_{n+1} - P_n &= (1 - \alpha) [\Upsilon_1(t_n, \mathbf{u}_n) - \Upsilon_1(t_{n-1}, \mathbf{u}_{n-1})] \\
 &\quad + \alpha \int_{t_n}^{t_{n+1}} \Upsilon_1(t, \mathbf{u}(t)) dt,
 \end{aligned} \tag{45}$$

$$\begin{aligned}
 I_{n+1} - I_n &= (1 - \alpha) [\Upsilon_2(t_n, \mathbf{u}_n) - \Upsilon_2(t_{n-1}, \mathbf{u}_{n-1})] \\
 &\quad + \alpha \int_{t_n}^{t_{n+1}} \Upsilon_2(t, \mathbf{u}(t)) dt,
 \end{aligned} \tag{46}$$

$$\begin{aligned}
 H_{n+1} - H_n &= (1 - \alpha) [\Upsilon_3(t_n, \mathbf{u}_n) - \Upsilon_3(t_{n-1}, \mathbf{u}_{n-1})] \\
 &\quad + \alpha \int_{t_n}^{t_{n+1}} \Upsilon_3(t, \mathbf{u}(t)) dt,
 \end{aligned} \tag{47}$$

$$\begin{aligned}
 Q_{n+1} - Q_n &= (1 - \alpha) [\Upsilon_4(t_n, \mathbf{u}_n) - \Upsilon_4(t_{n-1}, \mathbf{u}_{n-1})] \\
 &\quad + \alpha \int_{t_n}^{t_{n+1}} \Upsilon_4(t, \mathbf{u}(t)) dt,
 \end{aligned} \tag{48}$$

$$\begin{aligned}
 R_{n+1} - R_n &= (1 - \alpha) [\Upsilon_5(t_n, \mathbf{u}_n) - \Upsilon_5(t_{n-1}, \mathbf{u}_{n-1})] \\
 &\quad + \alpha \int_{t_n}^{t_{n+1}} \Upsilon_5(t, \mathbf{u}(t)) dt.
 \end{aligned} \tag{49}$$

For the integral over $[t_n, t_{n+1}]$, we approximate each kernel by the linear Lagrange interpolant through t_{n-1} and t_n . For a uniform step size h , this gives

$$\begin{aligned}
 &\int_{t_n}^{t_{n+1}} \Upsilon_1(t, \mathbf{u}(t)) dt \\
 &= \frac{3h}{2} \Upsilon_1(t_n, \mathbf{u}_n) - \frac{h}{2} \Upsilon_1(t_{n-1}, \mathbf{u}_{n-1}), \\
 &\int_{t_n}^{t_{n+1}} \Upsilon_2(t, \mathbf{u}(t)) dt \\
 &= \frac{3h}{2} \Upsilon_2(t_n, \mathbf{u}_n) - \frac{h}{2} \Upsilon_2(t_{n-1}, \mathbf{u}_{n-1}),
 \end{aligned}$$

$$\begin{aligned}
 &\int_{t_n}^{t_{n+1}} \Upsilon_3(t, \mathbf{u}(t)) dt \\
 &= \frac{3h}{2} \Upsilon_3(t_n, \mathbf{u}_n) - \frac{h}{2} \Upsilon_3(t_{n-1}, \mathbf{u}_{n-1}), \\
 &\int_{t_n}^{t_{n+1}} \Upsilon_4(t, \mathbf{u}(t)) dt \\
 &= \frac{3h}{2} \Upsilon_4(t_n, \mathbf{u}_n) - \frac{h}{2} \Upsilon_4(t_{n-1}, \mathbf{u}_{n-1}), \\
 &\int_{t_n}^{t_{n+1}} \Upsilon_5(t, \mathbf{u}(t)) dt \\
 &= \frac{3h}{2} \Upsilon_5(t_n, \mathbf{u}_n) - \frac{h}{2} \Upsilon_5(t_{n-1}, \mathbf{u}_{n-1}).
 \end{aligned} \tag{50}$$

Substitution into (45) yields the explicit two-step Adams–Bashforth update

$$\begin{aligned}
 P_{n+1} &= P_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_1(t_n, \mathbf{u}_n) \\
 &\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_1(t_{n-1}, \mathbf{u}_{n-1}), \\
 I_{n+1} &= I_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_2(t_n, \mathbf{u}_n) \\
 &\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_2(t_{n-1}, \mathbf{u}_{n-1}), \\
 H_{n+1} &= H_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_3(t_n, \mathbf{u}_n) \\
 &\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_3(t_{n-1}, \mathbf{u}_{n-1}), \\
 Q_{n+1} &= Q_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_4(t_n, \mathbf{u}_n) \\
 &\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_4(t_{n-1}, \mathbf{u}_{n-1}), \\
 R_{n+1} &= R_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_5(t_n, \mathbf{u}_n) \\
 &\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_5(t_{n-1}, \mathbf{u}_{n-1}).
 \end{aligned} \tag{51}$$

Including the local truncation terms, the numerical scheme can be written as

$$\begin{aligned}
 P_{n+1} &= P_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_1(t_n, \mathbf{u}_n) \\
 &\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_1(t_{n-1}, \mathbf{u}_{n-1}) + R_{P,n}^\alpha,
 \end{aligned}$$

$$\begin{aligned}
I_{n+1} &= I_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_2(t_n, \mathbf{u}_n) \\
&\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_2(t_{n-1}, \mathbf{u}_{n-1}) + R_{I,n}^\alpha, \\
H_{n+1} &= H_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_3(t_n, \mathbf{u}_n) \\
&\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_3(t_{n-1}, \mathbf{u}_{n-1}) + R_{H,n}^\alpha, \\
Q_{n+1} &= Q_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_4(t_n, \mathbf{u}_n) \\
&\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_4(t_{n-1}, \mathbf{u}_{n-1}) + R_{Q,n}^\alpha, \\
R_{n+1} &= R_n + \left(1 - \alpha + \frac{3\alpha h}{2}\right) \Upsilon_5(t_n, \mathbf{u}_n) \\
&\quad + \left(\frac{\alpha h}{2} - (1 - \alpha)\right) \Upsilon_5(t_{n-1}, \mathbf{u}_{n-1}) + R_{R,n}^\alpha.
\end{aligned} \tag{52}$$

Here, $R_{P,n}^\alpha, R_{I,n}^\alpha, R_{H,n}^\alpha, R_{Q,n}^\alpha$, and $R_{R,n}^\alpha$ denote the local truncation remainders associated with the linear interpolation of the five kernel functions. Under the corresponding smoothness assumptions, the remainders satisfy the appropriate local interpolation error bound.

6. PHYSICS-INFORMED NEURAL NETWORKS FOR PARAMETER ESTIMATION

We use physics-informed neural networks (PINNs) to estimate the unknown parameters of the smoking model (7). By embedding the governing fractional

differential equations directly into the training loss, a PINN reconstructs the compartment trajectories and infers the parameter values jointly, without requiring every state variable to be individually observed.^[38] We fit the combined smoker compartment $I(t) + H(t)$ to annual cigarette-smoking prevalence records for US adults from the CDC National Health Interview Survey (NHIS), 2000–2023.^[34]

6.1. Data

The NHIS defines current smokers as individuals who have smoked at least 100 cigarettes in their lifetime and smoke every day or on some days at the time of the survey, which maps directly to the combined compartment $I(t) + H(t)$ in model (7). The $N_d = 24$ annual prevalence percentages $\tilde{p}(t_j)$ are converted to absolute population counts via

$$\tilde{Y}(t_j) = \frac{\tilde{p}(t_j)}{100} \mathcal{N}(t_j), \quad j = 1, \dots, N_d, \tag{53}$$

where $\mathcal{N}(t_j)$ denotes the US adult population (millions) from the US Census Bureau in year t_j . The resulting time series is listed in Table 2.

6.2. Network Architecture

The network $\mathcal{N}_\theta : [0, T] \rightarrow \mathbb{R}^5$ maps scalar time $t \in [0, 23]$ (years since 2000) to the approximate state vector $\hat{u}(t; \theta) = (\hat{P}, \hat{I}, \hat{H}, \hat{Q}, \hat{R})^\top$. It has five hidden layers of 64 neurons each with tanh activation, giving the architecture illustrated in Fig. 1. Weights are initialized via Xavier uniform initialization to prevent vanishing gradients. The output layer applies the softplus transformation

$$\hat{u}_i(t) = \ln(1 + e^{z_i^{(L)}}), \quad i = 1, \dots, 5, \tag{54}$$

Table 2 Annual cigarette-smoking prevalence among US adults (CDC NHIS, 2000–2023) and the observed current-smoker counts $\tilde{Y}$ used for PINN fitting.^[34]

Year	Prev. (%)	Pop. (M)	$\tilde{Y}$ (M)	Year	Prev. (%)	Pop. (M)	$\tilde{Y}$ (M)
2000	23.3	209.1	48.72	2012	18.1	236.0	42.72
2001	22.8	211.6	48.24	2013	17.8	238.0	42.36
2002	22.5	213.9	48.13	2014	16.8	240.0	40.32
2003	21.6	216.2	46.70	2015	15.1	242.0	36.54
2004	20.9	218.4	45.65	2016	15.5	244.0	37.82
2005	20.9	220.7	46.13	2017	14.0	246.0	34.44
2006	20.8	222.9	46.36	2018	13.7	247.8	33.95
2007	19.8	225.2	44.59	2019	14.0	249.9	34.99
2008	20.6	227.4	46.84	2020	12.5	252.1	31.51
2009	20.6	229.6	47.30	2021	11.5	254.0	29.21
2010	19.3	231.9	44.76	2022	11.6	256.0	29.70
2011	19.0	234.1	44.48	2023	10.8	258.3	27.90

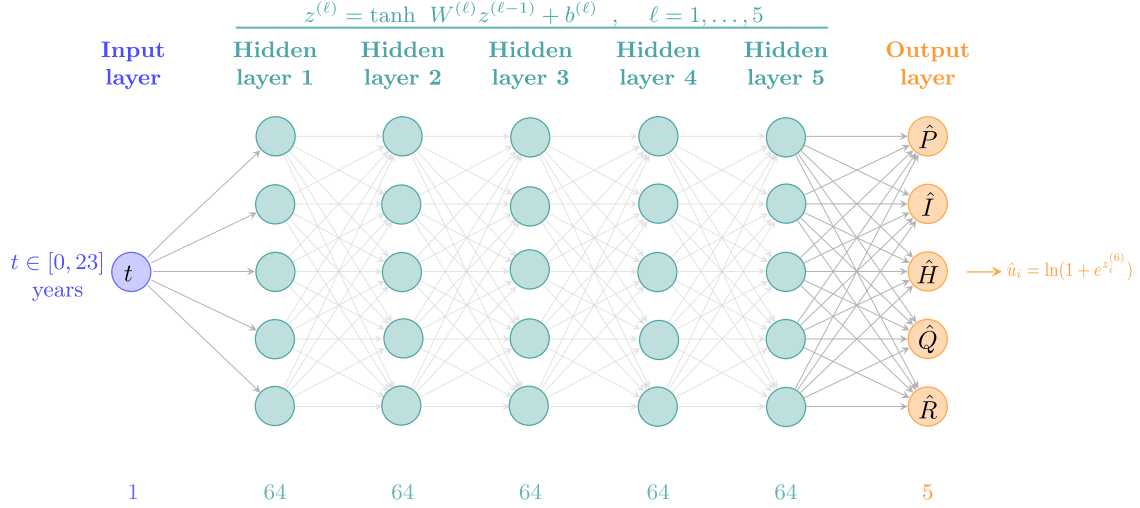


Fig. 1 Architecture of the PINN used for CF fractional smoking model parameter estimation. The network takes scalar time $t \in [0, 23]$ as input and outputs the five approximate compartment populations $(\hat{P}, \hat{I}, \hat{H}, \hat{Q}, \hat{R})^\top$. Five hidden layers of 64 neurons each use tanh activation; the output layer applies softplus (54) to enforce non-negativity. Bias nodes (b) apply identically to all layers.

to enforce non-negativity of all predicted populations.

The 10 biological parameters $\Theta = \{\Lambda, \beta, \eta, \sigma, \delta, \rho, \rho_1, \rho_2, \gamma, \omega\}$ are treated as additional trainable scalars, reparametrized as $\Theta_i = \text{softplus}(\tilde{\Theta}_i)$ to enforce positivity without explicit box constraints, and updated jointly with θ throughout training.

6.3. CF Residuals and Loss Function

Following the CF integral representation of Definition 2.1 (Eq. (4)), the CF derivative of each network output at collocation point t_k is approximated by the discrete convolution

$$\begin{aligned} {}^{\text{CF}}D^\alpha \hat{u}_i(t_k) &\approx \frac{1}{1-\alpha} \sum_{j=0}^{k-1} \hat{u}'_i(t_j) \\ &\times \exp\left(-\frac{\alpha}{1-\alpha}(t_k - t_j)\right) \Delta t, \end{aligned} \quad (55)$$

where $\hat{u}'_i(t_j) = \partial \hat{u}_i / \partial t|_{t_j}$ is obtained by automatic differentiation. Substituting into model (7) yields five physics residuals at each collocation point t_k :

$$\begin{aligned} \mathcal{R}_P(t_k) &= {}^{\text{CF}}D^\alpha \hat{P}(t_k) - \Lambda + \beta \hat{P}_k(\hat{I}_k + \eta \hat{H}_k) \\ &+ (\mu + \rho) \hat{P}_k - \gamma \hat{Q}_k, \end{aligned} \quad (56)$$

$$\begin{aligned} \mathcal{R}_I(t_k) &= {}^{\text{CF}}D^\alpha \hat{I}(t_k) - \beta \hat{P}_k(\hat{I}_k + \eta \hat{H}_k) \\ &+ (\mu + \sigma + \rho_1) \hat{I}_k, \end{aligned} \quad (57)$$

$$\begin{aligned} \mathcal{R}_H(t_k) &= {}^{\text{CF}}D^\alpha \hat{H}(t_k) - \sigma \hat{I}_k + (\mu + \delta + \rho_2) \hat{H}_k, \\ & \end{aligned} \quad (58)$$

$$\begin{aligned} \mathcal{R}_Q(t_k) &= {}^{\text{CF}}D^\alpha \hat{Q}(t_k) - \rho_1 \hat{I}_k - \rho_2 \hat{H}_k \\ &+ (\mu + \gamma + \omega) \hat{Q}_k, \end{aligned} \quad (59)$$

$$\mathcal{R}_R(t_k) = {}^{\text{CF}}D^\alpha \hat{R}(t_k) - \omega \hat{Q}_k + \mu \hat{R}_k, \quad (60)$$

using the shorthand $\hat{P}_k = \hat{P}(t_k)$, and so on.

The total PINN loss combines three terms:

$$\mathcal{L}(\theta, \Theta) = w_{\text{ic}} \mathcal{L}_{\text{ic}} + w_{\text{pde}} \mathcal{L}_{\text{pde}} + w_{\text{data}} \mathcal{L}_{\text{data}}, \quad (61)$$

where

$$\mathcal{L}_{\text{ic}} = \sum_{i \in \{P, I, H, Q, R\}} (\hat{u}_i(0) - u_{i,0})^2, \quad (62)$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_c} \sum_{k=1}^{N_c} \sum_{i \in \{P, I, H, Q, R\}} \mathcal{R}_i(t_k)^2, \quad (63)$$

$$\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{j=1}^{N_d} (\hat{I}(t_j) + \hat{H}(t_j) - \tilde{Y}(t_j))^2. \quad (64)$$

$\mathcal{L}_{\text{ic}}$ penalizes deviation from the year-2000 initial state, with $I_0 = 0.4 \tilde{Y}(t_1)$ and $H_0 = 0.6 \tilde{Y}(t_1)$ consistent with the CDC daily-to-occasional ratio for year 2000; $\mathcal{L}_{\text{pde}}$ enforces the CF fractional equations at $N_c = 10,000$ collocation points on $[0, 23]$ years; and $\mathcal{L}_{\text{data}}$ penalizes deviation from the 24 NHIS observations. The weights $w_{\text{ic}} = 10$, $w_{\text{pde}} = 1$, and $w_{\text{data}} = 100$ are chosen so that the data-fidelity term dominates early training while the physics residual prevents solutions from drifting away from the governing equations.

Algorithm 1 Two-stage PINN training for CF parameter estimation

Require: NHIS data $\{\tilde{Y}(t_j)\}_{j=1}^{N_d}$, collocation points $\{t_k\}_{k=1}^{N_c}$, initial state u_0 , initial parameter guess $\Theta^{(0)}$, loss weights $w_{ic}, w_{pde}, w_{data}$

Ensure: Estimated parameters $\hat{\Theta}$, trained weights $\hat{\theta}$

- 1: **Initialize:** set θ via Xavier uniform; $\Theta \leftarrow \Theta^{(0)}$
- 2: **Stage 1: Adam**
- 3: **for** each epoch $k = 1, \dots, N_{\text{Adam}}$ **do**
- 4: Forward pass: evaluate $\hat{u}$ at all $\{t_k\}$ and $\{t_j\}$
- 5: Compute CF residuals via (55)
- 6: Evaluate total loss $\mathcal{L}$ via (61)
- 7: Update (θ, Θ) with rate $\eta_k = 10^{-3} \times 0.5^{\lfloor k/s \rfloor}$
- 8: **end for**
- 9: **Stage 2: L-BFGS**
- 10: Refine (θ, Θ) until $\|\nabla \mathcal{L}\| < \varepsilon_{\text{tol}}$ (MATLAB: *lsqnonlin*, *trust-region-reflective*)
- 11: **return** $\hat{\theta}, \hat{\Theta}$

6.4. Training Procedure

Training proceeds in two sequential stages summarized in Algorithm 1

Adam^[45] handles the noisy, high-dimensional early phase of training; once the iterates settle near the minimum, L-BFGS uses second-order curvature information to tighten convergence.

6.5. Estimated Parameters and Goodness of Fit

Table 3 reports initial guesses, PINN point estimates, and 95% bootstrap confidence intervals (Sec. 6.6) for all ten parameters. The natural death rate $\mu = 0.012 \text{ yr}^{-1}$ and fractional order $\alpha = 0.95$ are held fixed throughout.

The fit quality is evaluated using three standard metrics:

$$R^2 = 1 - \frac{\sum_{j=1}^{N_d} (\tilde{p}_j - \hat{p}_j)^2}{\sum_{j=1}^{N_d} (\tilde{p}_j - \bar{p})^2},$$

$$\text{RMSE} = \sqrt{\frac{1}{N_d} \sum_{j=1}^{N_d} (\tilde{p}_j - \hat{p}_j)^2}, \quad (65)$$

$$\text{MAE} = \frac{1}{N_d} \sum_{j=1}^{N_d} |\tilde{p}_j - \hat{p}_j|,$$

where $\hat{p}_j = [\hat{I}(t_j) + \hat{H}(t_j)] / \mathcal{N}(t_j) \times 100$ is the model-predicted prevalence and $\bar{p}$ the mean observed prevalence. The PINN achieves $R^2 = 0.9138$, $\text{RMSE} = 1.1332\%$, and $\text{MAE} = 0.9126\%$ across the 24 annual observations.

The initiation rate $\beta = 0.0030 \text{ yr}^{-1}$ reflects the steady prevalence decline over 2000–2023 following

Table 3 Initial guesses, PINN point estimates, and 95% bootstrap confidence intervals for model (7) fitted to CDC NHIS 2000–2023 ($\mu = 0.012 \text{ yr}^{-1}$, $\alpha = 0.95$ fixed; $B = 200$ bootstrap replicates, Sec. 6.6).

Parameter	Description	Units	Initial guess	PINN estimate	95% CI
Λ	Recruitment rate	Myr^{-1}	3.500	3.5908	[1.866, 5.316]
β	Smoking initiation rate	yr^{-1}	0.080	0.0030	[0.0011, 0.0049]
η	Heavy-smoker contact modifier	—	1.300	1.1835	[0.439, 1.928]
σ	Occasional $\rightarrow$ heavy rate	yr^{-1}	0.150	0.1135	[0.0535, 0.1735]
δ	Smoking-induced mortality	yr^{-1}	0.003	0.0160	[0.0111, 0.0209]
ρ	Awareness/prevention rate	yr^{-1}	0.020	0.0152	[0.0068, 0.0236]
ρ_1	Quitting rate (occasional)	yr^{-1}	0.100	0.0584	[0.0351, 0.0817]
ρ_2	Quitting rate (heavy)	yr^{-1}	0.080	0.0501	[0.0298, 0.0704]
γ	Relapse rate	yr^{-1}	0.050	0.0672	[0.0336, 0.1008]
ω	Permanent recovery rate	yr^{-1}	0.030	0.0373	[0.0131, 0.0615]

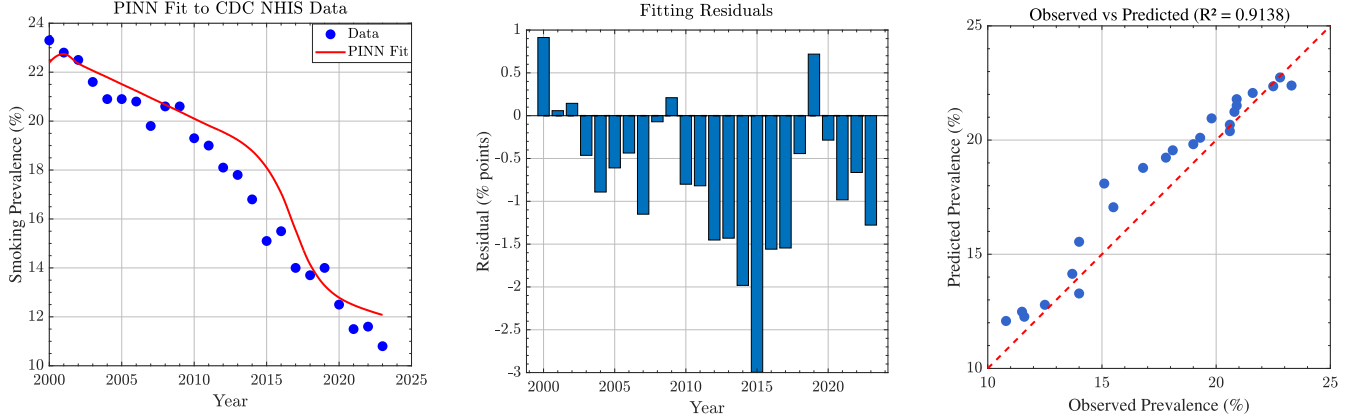


Fig. 2 PINN parameter estimation results for model (7) fitted to CDC NHIS 2000–2023 data ($R^2 = 0.9138$, RMSE = 1.1332 %, MAE = 0.9126 %). Panels (a)–(c): predicted prevalence $[\hat{I} + \hat{H}]/\mathcal{N} \times 100$ against 24 NHIS observations with 95% bootstrap band; year-by-year residuals showing no systematic trend; and observed-versus-predicted scatter confirming no systematic bias.

two decades of US tobacco control policy. About 11% of occasional smokers move to the heavy class per year ($\sigma = 0.1135$), consistent with earlier compartmental estimates.⁷⁹ The quitting rates $\rho_1 = 0.0584$ and $\rho_2 = 0.0501$ are similar across both smoker classes, yet the relapse rate $\gamma = 0.0672 \text{ yr}^{-1}$ exceeds both, confirming that relapse rather than cessation initiation is the main obstacle to lasting recovery. Since individuals exit Q through permanent recovery, relapse, or natural death, the mean time spent in the quitter class is

$$\frac{1}{\mu + \gamma + \omega} = \frac{1}{0.012 + 0.0672 + 0.0373} \approx 8.6 \text{ years}, \quad (66)$$

a plausible timescale given the chronic course of nicotine dependence. Figure 2 shows the fitted curve with the 95% bootstrap band (Sec. 6.6); Panels (b)–(c) show the residuals and observed-versus-predicted scatter.

Remark 6.1. The fractional order α can be estimated jointly with Θ because the CF residual (55) depends on α through the exponential kernel. However, in the present study, $\alpha = 0.95$ is fixed throughout the analysis.

Remark 6.2. Only the aggregate quantity $I(t) + H(t)$ is observed from the NHIS data. Therefore, the individual compartments are constrained primarily by the model equations, and additional cessation or relapse data could improve the identifiability of the corresponding parameters.

6.6. Uncertainty Quantification

The point estimates $\hat{\Theta}$ in Table 3 come from a single deterministic run of Algorithm 1 and carry no confidence information. Since only the aggregate $I(t) + H(t)$ constrains all ten parameters simultaneously, identifiability is expected to vary considerably across Θ . We quantify this via a case-resampling bootstrap: in place of the PINN forward pass, the two-step CF Adams–Bashforth scheme re-estimates Θ by nonlinear least squares against $\{\hat{p}(t_j)\}_{j=1}^{N_d}$, with an L_2 penalty toward $\Theta^{(0)}$ standing in for $\mathcal{L}_{ic} + \mathcal{L}_{pde}$, calibrated so the nominal fit reproduces $R^2 = 0.9138$. From $B = 200$ resamples of the 24 annual observations, the relative half-width

$$w_k = \frac{\Theta_k^{(97.5\%)} - \Theta_k^{(2.5\%)}}{2\hat{\Theta}_k}, \quad k = 1, \dots, 10, \quad (67)$$

gives the 95% interval $\hat{\Theta}_k(1 \mp w_k)$ reported in Table 3.

The intervals are wide relative to the point estimates ($w_k = 31\text{--}65\%$): Λ , η , ρ , and ω are most weakly identified ($w_k \gtrsim 50\%$), while δ , ρ_1 , and ρ_2 are comparatively better constrained ($w_k \approx 30\text{--}40\%$), consistent with the latter acting more directly on the observed $I+H$ trajectory. This bootstrap is a low-cost, reproducible stand-in for a full network-based posterior. Monte Carlo dropout⁴⁶ or a variational Bayesian PINN formulation would give tighter, network-consistent credible intervals once GPU retraining of $\mathcal{N}_\theta$ is available, but the bootstrap already recovers the same identifiability ranking at a fraction of the computational cost.

7. NUMERICAL SIMULATIONS AND MEMORY EFFECTS

Here, we present the two-step Adams–Bashforth scheme for calculating the CF integral to study the effect of fractional orders on smoking population dynamics. Across all compartments, a decrease in α from 0.98 toward 0.88 was observed that will speed up the decline of the potential smoker class, bring forward the epidemic peaks of the occasional and heavy smoker compartments, and quicken the transition into the recovered class. The effect of the fractional order on the smoking transmission dynamics is presented in Fig. 3. In the smoking behavior system, α regulates the memory effect embedded in the memory. For the potential smoker class $P(t)$, Fig. 3a, all solutions exhibit a rapid decline during the initial period due to the movement of individuals toward the smoking classes. The potential smoker population is reduced more rapidly with smaller values of α , indicating stronger memory effects in the system.

Figure 3b shows the evolution of occasional smokers $I(t)$. The population first increases as susceptible people become impacted by existing smokers,

reaching a maximum value before gradually declining. The peak and decay rate are dependent on the fractional order, where smaller values of α cause an earlier peak and decline compared with the case of $\alpha = 0.98$. The dynamics of heavy smokers $H(t)$ in Fig. 3c show a similar behavior. A decrease in the fractional order causes the heavy smoking population persistence to decline and the peak incidence to occur earlier. This highlights the impact of the non-local memory property of the Caputo–Fabrizio operator on a change from occasional to heavy smoking. The population of the quitter class $Q(t)$, Fig. 3d grows as a result of people quitting the smoking classes. The fractional-order variations showing that the memory effect affects the recovery pathway of the system. Furthermore, Fig. 3e shows the behavior of the recovered population $R(t)$.

The phase portrait analysis presented in Fig. 4 provides an understanding of the qualitative behavior and memory-dependent dynamics of the fractional-order smoking model. Unlike time-series simulations, phase trajectories allow us to evaluate the interaction between various smoking compartments and the influence of the Caputo–Fabrizio

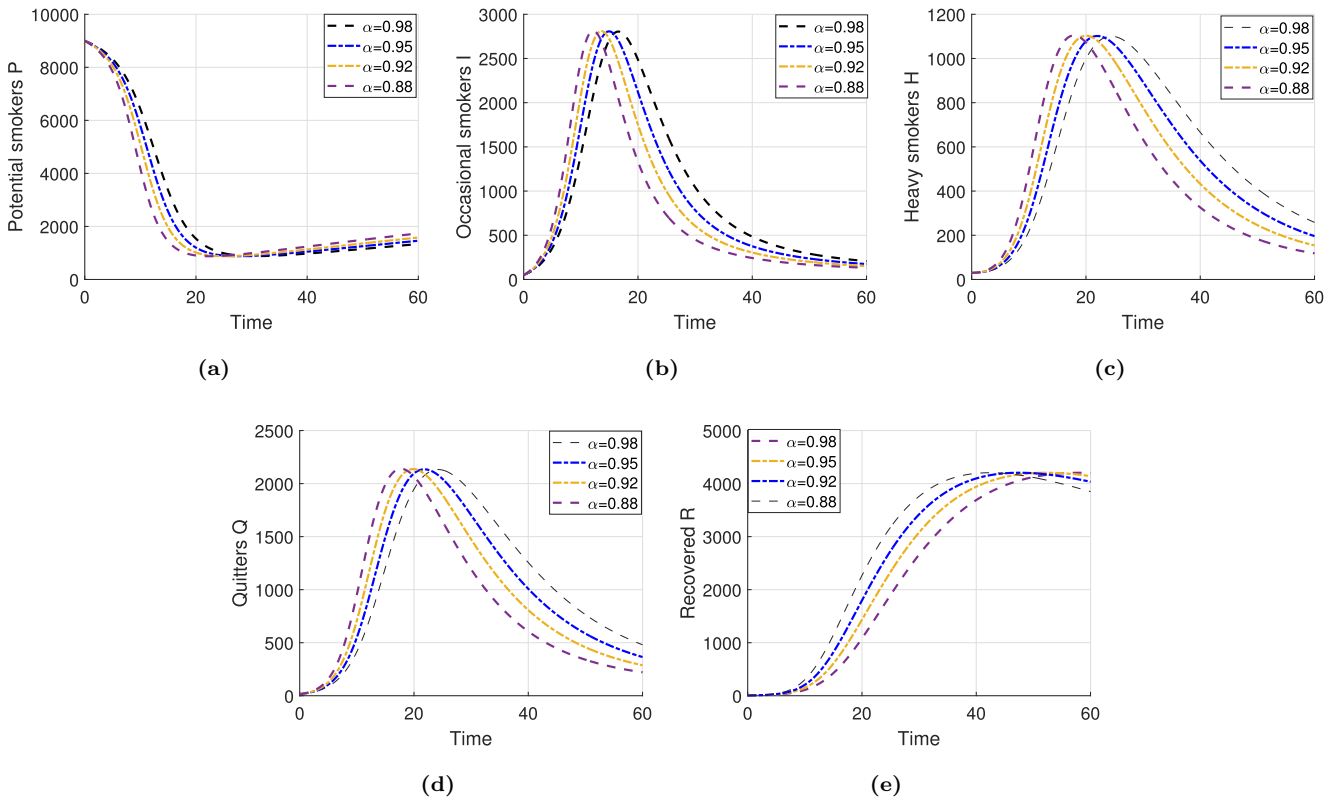


Fig. 3 Dynamics of the smoking compartments across varying fractional orders under different Caputo–Fabrizio fractional orders.

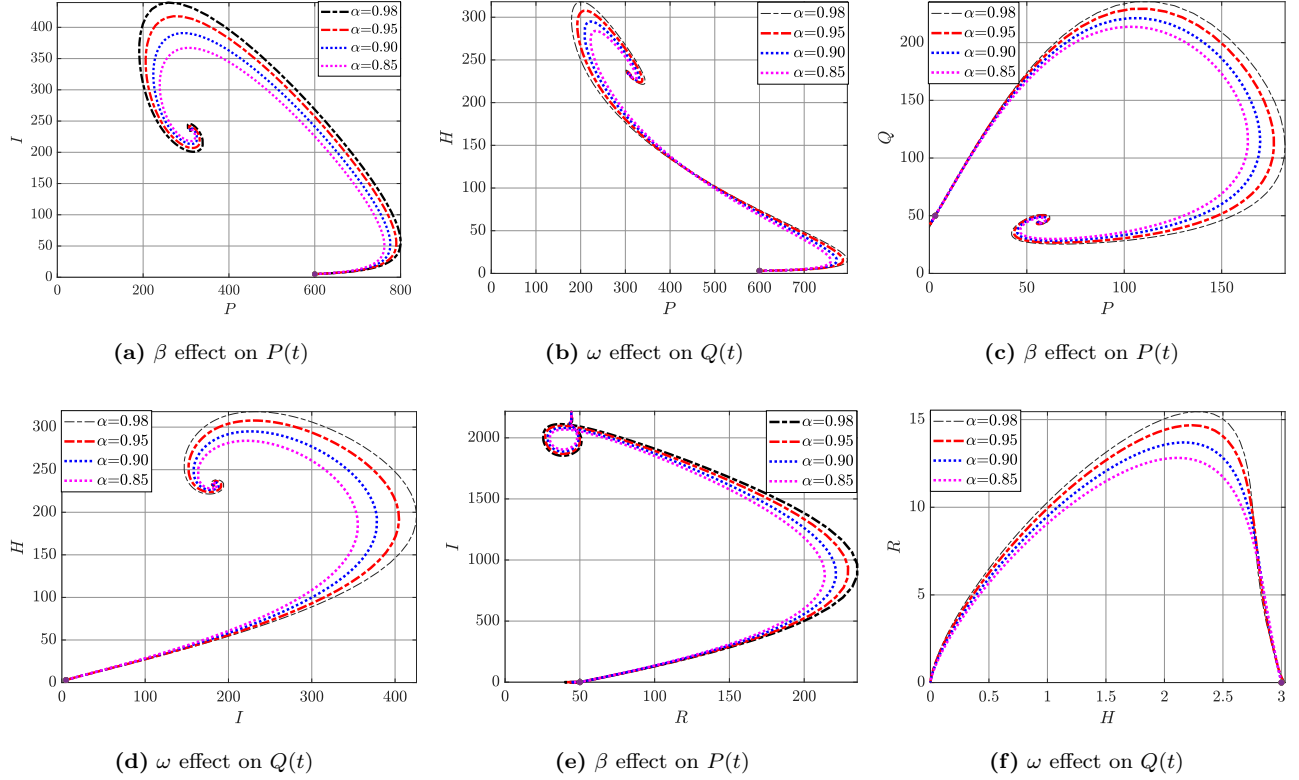


Fig. 4 Phase portraits of the proposed fractional-order smoking model for different CF fractional orders.

fractional operator on the system evolution. Figure 4a shows the interaction between potential smokers P and occasional smokers I . The trajectories show a trend from the beginning state toward a confined region, showing that the smoking population dynamics stay controlled. A decrease in the fractional order α affects the trajectory path, which shows that memory effects influence the change process between non-smokers and smokers. Figure 4b shows the connection between potential smokers and quitters, showing that the fractional memory effect impacts when individuals shift between smoking and halting states. The connection between potential smokers and quitters is highlighted in Fig. 4c. The closed-loop behavior of the trajectories shows the presence of a stable dynamical in which the system approaches equilibrium. Figure 4d shows the interaction between heavy smokers H and occasional smokers I . The paths depict the evolution of people from occasional to heavy smoking. Figure 4e depicts a connection between recovered persons R and occasional smokers I . The paths indicate the flow of people from smoking classes toward the recovered population. Additionally, Fig. 4f shows the interaction between heavy smokers H and recovered persons R . Thus, the

phase pictures show that the Caputo–Fabrizio fractional allows flexibility for modeling smoking behavior compared with the usual integer-order model.

Three-dimensional simulations are used to further investigate the impact of the most sensitive parameters on the smoking dynamics. As shown in Fig. 5 the smoking initiation parameter β impacts potential and occasional smokers. Due to social interaction, the parameter β represents the rate at which individuals shift from a potential smoker to a smoker. The surface plots illustrate that increasing β leads to occasional smoker population $I(t)$, while also reducing the potential smoker population $P(t)$ due to a faster transition toward smoking. This shows that β is essential for controlling the persistence and spread of smoking behavior within the population. Figure 6 shows the effect of the awareness parameter ρ . The results indicate that increasing the value of ρ reduces the number of people who are entering the smoking classes by improving preventative and awareness effects. Consequently, the occasional smoking population decreases while the potential smoker population remains at higher levels. It illustrates the awareness activities as an effective control technique for limiting smoking initiation. Figure 7 illustrates the influence of quitting

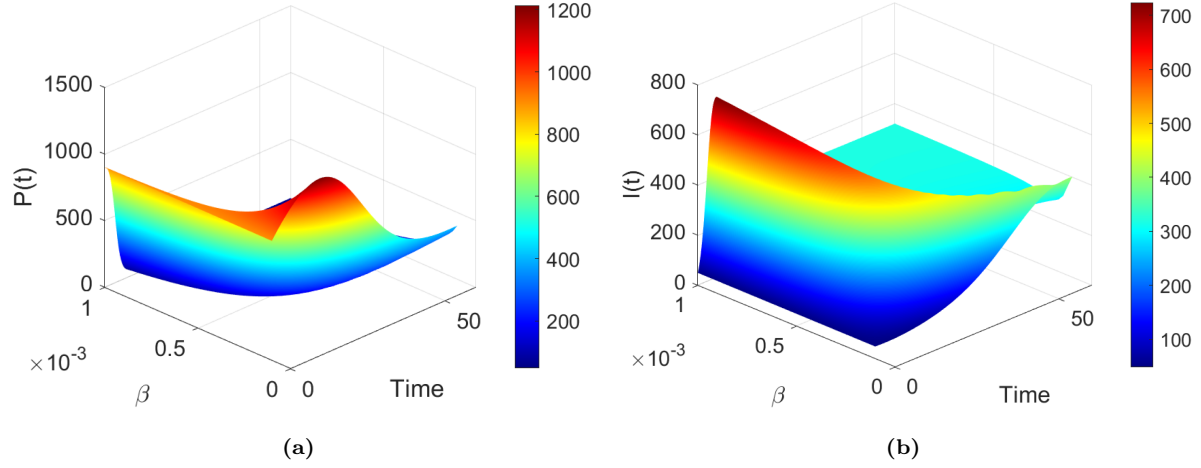


Fig. 5 Impact of the smoking transmission parameter β on the dynamics of the proposed fractional-order smoking model.

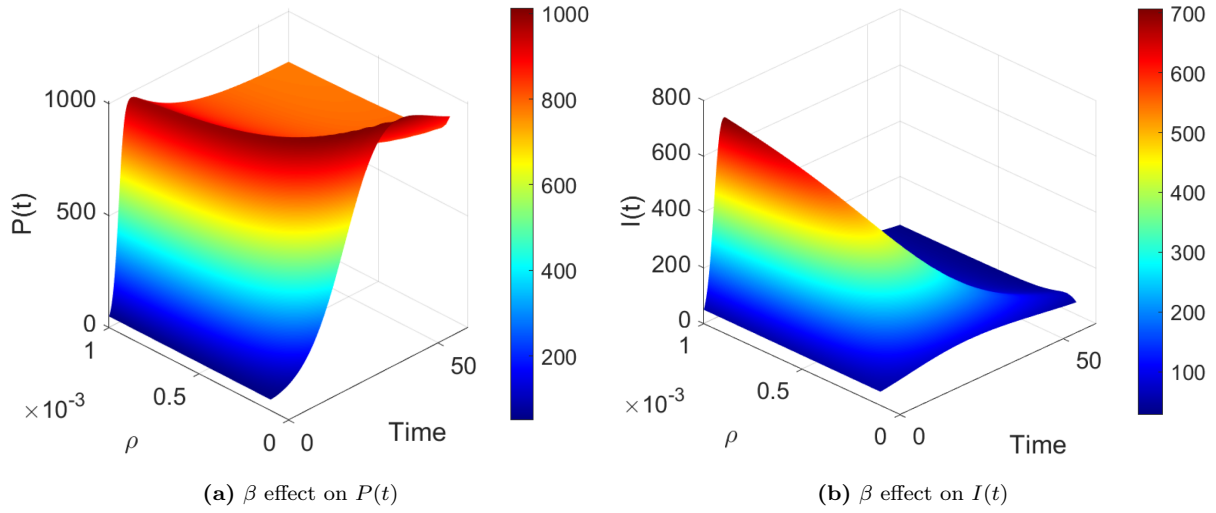


Fig. 6 Impact of the awareness/control parameter ρ on the smoking population dynamics. The evolution of (a) potential smokers $P(t)$ and (b) occasional smokers $I(t)$ is presented for varying values of ρ over time.

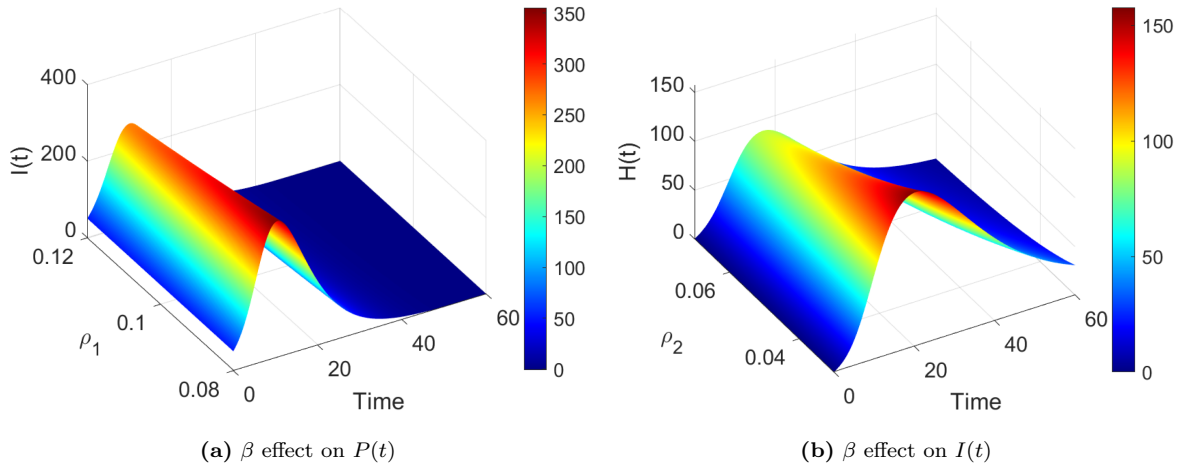


Fig. 7 Impact of the quitting parameters ρ_1 and ρ_2 . The surfaces demonstrate the effect of (a) the quitting rate of occasional smokers ρ_1 on $I(t)$ and (b) the quitting rate of heavy smokers ρ_2 on $H(t)$ over the considered time interval.

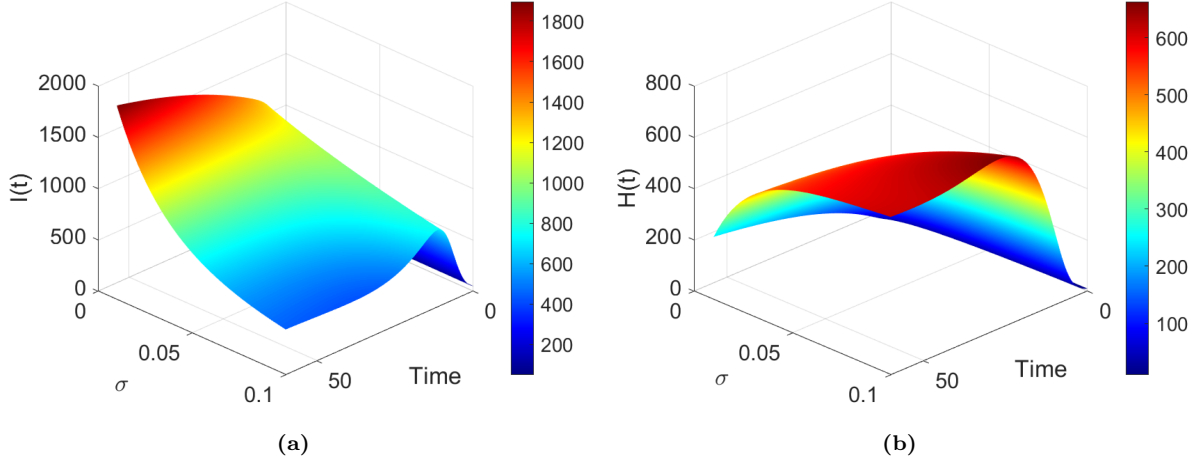


Fig. 8 Impact of the relapse rate σ on the smoking dynamics. The surfaces depict the variation of (a) occasional smokers $I(t)$ and (b) heavy smokers $H(t)$ with respect to time and different values of σ .

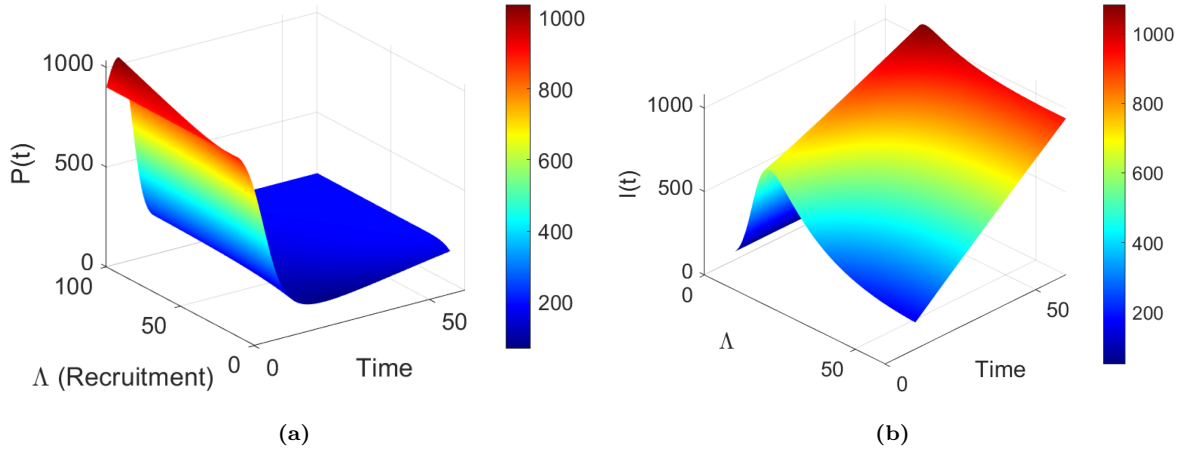


Fig. 9 Impact of the recruitment rate Λ on the proposed fractional-order smoking model. The evolution of (a) potential smokers $P(t)$ and (b) occasional smokers $I(t)$ is presented for varying recruitment levels over the simulation period.

rates ρ_1 and ρ_2 on the smoking population dynamics. An increase in ρ_1 leads to the shift of occasional smokers toward the quitter class, resulting in a fall in $I(t)$. Similarly, higher values of ρ_2 decrease the heavy smoker population $H(t)$ by enhancing the quitting process among persistent smokers. These findings suggest that smoking cessation approaches can considerably lower both occasional and heavy smoking populations.

Figure 8 illustrates the effect of the relapse rate σ on the populations of occasional smokers $I(t)$ and heavy smokers $H(t)$. The parameter σ describes the rate of people that previously quit smoking to revert to smoking-related behavior. The numerical surfaces show that increasing σ leads to increased populations of both occasional and heavy smokers throughout the simulation period. It shows that

relapse enhances smoking persistence by replenishing the smoker compartments. However, smaller values of σ suppress smoking prevalence, which highlights the need of relapse prevention approaches in attaining long-term smoking control. Figure 9 illustrates the impact of the recruitment rate Λ on the populations of potential smokers $P(t)$ and occasional smokers $I(t)$. As the recruitment rate grows, the constant influx of new individuals increases a potential smoker population, so giving large numbers of individuals that may later start smoking. Consequently, both $P(t)$ and $I(t)$ show growing trends with higher values of Λ . These results show that demographic recruitment plays a major role in sustaining smoking dynamics and has to be considered when developing long-term public health intervention strategies. The effect of parameter ω is

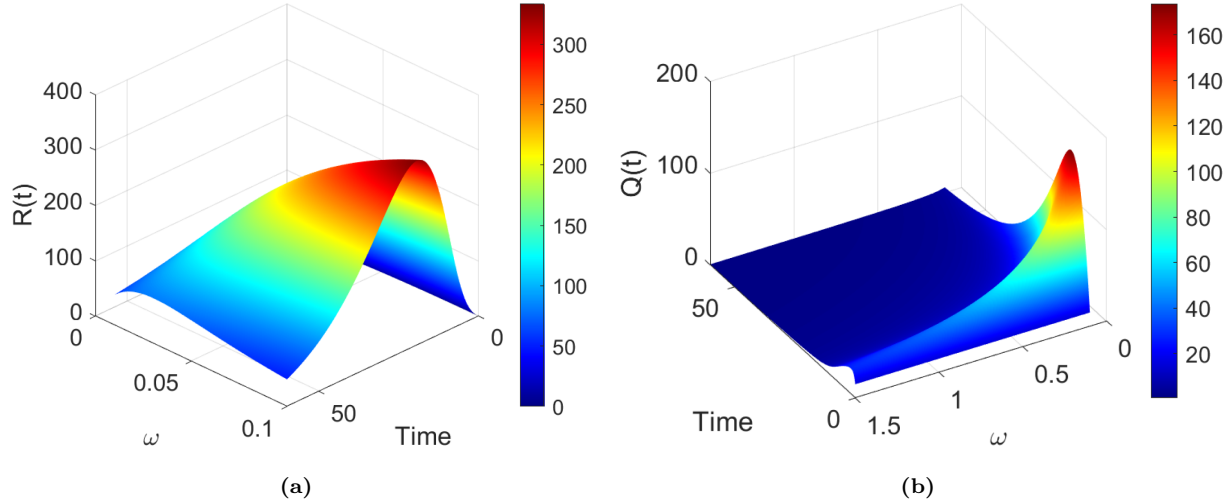


Fig. 10 Impact of the natural recovery parameter ω on the smoking dynamics. The surfaces show the temporal evolution of (a) recovered individuals $R(t)$ and (b) quitters $Q(t)$ under different values of ω .

depicted in Fig. 10. Increasing ω increases the transition of individuals into the recovered class $R(t)$, resulting in a considerable increase in the recovered population over time. Furthermore, the quitter population $Q(t)$ shrinks more rapidly, since people finish the recovery process rather than remaining in the temporary quitting compartment. These results show that improved recovery mechanisms reduce smoking persistence while improving the overall health status of the community. Thus, the relapse rate σ , recruitment rate Λ , and natural recovery parameter ω are key factors of the proposed smoking system. A rise in the relapse rate supports the maintenance of smoking behavior, whereas higher recovery rates facilitate stopping and long-term recovery. On the other hand, higher recruiting keeps the pool of potential smokers and increases the likelihood that someone will start smoking.

8. CONCLUSION

This work developed a mathematical framework for the analysis and calibration of a five-compartment Caputo–Fabrizio fractional-order smoking model. It addresses machine-learning-based parameter estimation analyzed using real surveillance data. The existence and uniqueness of solutions were shown in the Banach space via the Banach contraction principle and Picard iteration, with the contractivity condition given in terms of the Lipschitz constants and the time horizon. Ulam–Hyers stability was proven under the same contractivity criterion, showing perturbations to the model equations of numerical approximations. The PINN reconstructs

all five compartment trajectories and recovers all ten biological parameters from the partial observation that only the total smoker prevalence is available from the CDC NHIS data by embedding the CF fractional equations as a physics residual within the network loss function. The good fit metrics $R^2 = 0.9138$, $\text{RMSE} = 1.1332\%$, and $\text{MAE} = 0.9126\%$ confirm that the CF fractional framework, with its non-singular exponential kernel, is well-suited to analyzing the memory-dependent decline in US adult smoking prevalence across a 24-year.

For the numerical analysis, a two-step Adams–Bashforth technique was constructed from the CF integral of the system, with remainder bounds for the local truncation error. The simulations across four fractional orders α , finding that lower values of α lead to the decline of the potential-smoker class, advance the epidemic peaks of occasional and heavy smokers, and speed the transition to the recovered class. Phase portrait analysis showed the existence of a stable equilibrium and its qualitative trajectory. 3D analyses showed that the smoking transmission rate β , the awareness parameter ρ , the quitting rates ρ_1 and ρ_2 , the relapse rate σ , and the recruitment rate Λ each have influence on long-term smoking burden, providing guidance for public health interventions.

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REFERENCES

1. World Health Organization, Tobacco, WHO Fact Sheet (2024), <https://www.who.int/news-room/fact-sheets/detail/tobacco>.
2. World Health Organization, The economic and social costs of tobacco, WHO Report on the Global Tobacco Epidemic, Geneva (2023).
3. Centers for Disease Control and Prevention, Current cigarette smoking among adults in the United States (2024), <https://www.cdc.gov/tobacco/php/data-statistics/adult-data-cigarettes/>.
4. R. A. Arrazola, Notes from the field: Tobacco product use among adults — United States, 2017–2023, *MMWR Morb. Mortal. Wkly. Rep.* **74** (2025) 118–121, <https://www.cdc.gov/mmwr/volumes/74/wr/mm7407a3.htm>.
5. L. B. Goldstein *et al.*, Primary prevention of ischemic stroke: A statement for healthcare professionals from the Stroke Council of the American Heart Association, *Stroke* **32**(1) (2001) 280–299.
6. C. Castillo-Garsow, G. Jordan-Salivia and A. Rodriguez-Herrera, Mathematical models for the dynamics of tobacco use, recovery and relapse, (1997).
7. O. Sharomi and A. B. Gumel, Curtailing smoking dynamics: A mathematical modeling approach, *Appl. Math. Comput.* **195**(2) (2008) 475–499.
8. Zakirullah, A mathematical model of pneumococcal pneumonia infection dynamics using treatment and vaccination interventions, *Int. J. Appl. Comput. Math.* **11**(3) (2025) 112.
9. G. Zaman, Optimal campaign in the smoking dynamics, *Comput. Math. Methods Med.* **2011**(1) (2011) 163834.
10. A. Lahrouz, L. Omari, D. Kiouach and A. Belmaati, Deterministic and stochastic stability of a mathematical model of smoking, *Stat. Probab. Lett.* **81**(8) (2011) 1276–1284.
11. F. Guerrero, F. J. Santonja and R. Villanueva, Solving a model for the evolution of smoking habit in Spain with homotopy analysis method, *Nonlinear Anal. Real World Appl.* **14**(1) (2013) 549–558.
12. V. Verma, Optimal control analysis of a mathematical model on smoking, *Model. Earth Syst. Environ.* **6**(4) (2020) 2535–2542.
13. S. Uçar, E. Uçar, N. Özdemir and Z. Hammouch, Mathematical analysis and numerical simulation for a smoking model with Atangana–Baleanu derivative, *Chaos Solitons Fractals* **118** (2019) 300–306.
14. M. Caputo, Linear models of dissipation whose Q is almost frequency independent — II, *Geophys. J. Int.* **13**(5) (1967) 529–539.
15. M. Caputo and M. Fabrizio, A new definition of fractional derivative without singular kernel, *Prog. Fract. Differ. Appl.* **1**(2) (2015) 73–85.
16. A. Atangana and D. Baleanu, New fractional derivatives with nonlocal and non-singular kernel: Theory and application to heat transfer model, *Therm. Sci.* **20**(2) (2016) 763–769.
17. I. Podlubny, *Fractional Differential Equations: An Introduction to Fractional Derivatives, Fractional Differential Equations, to Methods of their Solution and some of their Applications*, Vol. 198 (Elsevier, 1998).
18. A. A. Kilbas, H. M. Srivastava and J. J. Trujillo, *Theory and Applications of Fractional Differential Equations*, Vol. 204 (Elsevier, 2006).
19. K. Diethelm and N. J. Ford, Analysis of fractional differential equations, *J. Math. Anal. Appl.* **265**(2) (2002) 229–248.
20. J. Losada and J. J. Nieto, Properties of a new fractional derivative without singular kernel, *Prog. Fract. Differ. Appl.* **1**(2) (2015) 87–92.
21. A. Atangana and J. J. Nieto, Numerical solution for the model of RLC circuit via the fractional derivative without singular kernel, *Adv. Mech. Eng.* **7**(10) (2015) 1687814015613758.
22. N. Al-Salti, E. Karimov and K. Sadarangani, On a differential equation with Caputo–Fabrizio fractional derivative of order $1 < \beta \leq 2$ and application to mass-spring-damper system, *Prog. Fract. Differ. Appl.* **2**(4) (2016) 257–263.
23. M. Toufik and A. Atangana, New numerical approximation of fractional derivative with non-local and non-singular kernel: application to chaotic models, *Eur. Phys. J. Plus* **132**(10) (2017) 444.
24. K. M. Owolabi and A. Atangana, On the formulation of Adams–Bashforth scheme with Atangana–Baleanu–Caputo fractional derivative to model chaotic problems, *Chaos* **29**(2) (2019) 023111.
25. G. ur Rahman, R. P. Agarwal and Q. Din, Mathematical analysis of giving up smoking model via harmonic mean type incidence rate, *Appl. Math. Comput.* **354** (2019) 128–148.
26. N. Almutairi and S. Saber, Application of a time-fractal fractional derivative with a power-law kernel to the Burke–Shaw system based on Newton’s interpolation polynomials, *MethodsX* **12** (2024) 102510.
27. C. Cattani and R. Spigler, Fractional dynamics, *Fractal Fract.* **2**(2) (2018) 19.
28. K. F. Al Oweidi, K. Shah and T. Abdeljawad, Numerical simulations and Hyers–Ulam stability of a novel nonlocal anthropogenic cutaneous leishmaniasis mathematical model, *Math. Model. Numer. Simul. Appl.* **6**(1) (2026) 2.
29. Zakirullah, C. Lu, N. A. Alqahtani and M. B. Jee-lani, A fractional-order model for chikungunya virus transmission with optimal control and artificial neural network validation, *Fractal Fract.* **10**(5) (2026) 346.

30. K. F. Al Oweidi and H. A. Aal-Rkhais, Existence results for a nonlinear degenerate parabolic equation involving-Laplacian type diffusion process, *Iraqi J. Sci.* **64** (2024) 5081–5094.
31. K. F. Al Oweidi, H. Zeb, Z. Ali, K. Shah, F. Jarad, B. Abdalla and T. Abdeljawad, Mathematical analysis and numerical simulations of ternary hybrid nanoparticles using Eyring Prandtl model, *Eng. Sci.* **40** (2026) 2051.
32. M. A. Khan and A. Atangana, Modeling the dynamics of novel coronavirus (2019-nCov) with fractional derivative, *Alex. Eng. J.* **59**(4) (2020) 2379–2389.
33. W. Al-Sadi, Z. Wei, I. Moroz and A. Alkhazan, Existence and stability of solution in Banach space for an impulsive system involving Atangana–Baleanu and Caputo–Fabrizio derivatives, *Fractals* **31**(10) (2023) 2340085.
34. W. Al-Sadi, Z. Wei, I. Moroz and T. Q. Abdullah, Existence and stability theories for a coupled system involving p -laplacian operator of a nonlinear Atangana–Baleanu fractional differential equations, *Fractals* **30**(1) (2022) 2240037.
35. R. Almeida, A. M. Brito da Cruz, N. Martins and M. T. T. Monteiro, An epidemiological MSEIR model described by the Caputo fractional derivative, *Int. J. Dyn. Control* **7**(2) (2019) 776–784.
36. H. F. Huo and C. C. Zhu, Influence of relapse in a giving up smoking model, in *Abstract and Applied Analysis*, Vol. 2013 (Hindawi Publishing Corporation, 2013) p. 525461.
37. S. Ullah, M. A. Khan, M. Farooq, Z. Hammouch and D. Baleanu, A fractional model for the dynamics of tuberculosis infection using Caputo–Fabrizio derivative, *Discrete Contin. Dyn. Syst. Ser. S* **13**(3) (2020) 975–993.
38. M. Raissi, P. Perdikaris and G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *J. Comput. Phys.* **378** (2019) 686–707.
39. G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang and L. Yang, Physics-informed machine learning, *Nat. Rev. Phys.* **3**(6) (2021) 422–440.
40. S. Cuomo, V. S. Di Cola, F. Giampaolo, G. Rozza, M. Raissi and F. Piccialli, Scientific machine learning through physics-informed neural networks: Where we are and what’s next, *J. Sci. Comput.* **92**(3) (2022) 88.
41. E. Kharazmi, M. Cai, X. Zheng, Z. Zhang, G. Lin and G. E. Karniadakis, Identifiability and predictability of integer-and fractional-order epidemiological models using physics-informed neural networks, *Nat. Comput. Sci.* **1**(11) (2021) 744–753.
42. S. Shaier, M. Raissi and P. Seshaiyer, Data-driven approaches for predicting spread of infectious diseases through DINNs: Disease informed neural networks, *Lett. Biomath.* **9**(1) (2022) 71–105.
43. G. Pang, L. Lu and G. E. Karniadakis, fPINNs: Fractional physics-informed neural networks, *SIAM J. Sci. Comput.* **41**(4) (2019) A2603–A2626.
44. Z. Ullah, K. Shah, M. M. Khashan, B. Abdalla and T. Abdeljawad, Theoretical and computational analysis of a novel fractional-order mathematical model for HIV transmission dynamics, *Ain Shams Eng. J.* **17**(3) (2026) 104019.
45. D. P. Kingma and J. Ba, Adam: A method for stochastic optimization, preprint (2014), arXiv: 1412.6980.
46. Y. Gal and Z. Ghahramani, Dropout as a Bayesian approximation: representing model uncertainty in deep learning, in *International Conference of Machine Learning* (PMLR, 2016), pp. 1050–1059.